

UNIT ONE: INTRODUCTION

1.1 PURPOSES OF SPECIAL NEEDS EDUCATION

- It **solves problems of exceptional children** in the regular classroom.
- It **creates awareness about problems of children with special needs among parents, teachers and the society at large. Hence, it brings about attitudinal changes.**
- It helps in **providing exceptional children with the necessary psychological, educational and social support that help them exploit and optimize their potentials.**
- It helps in **designing interventions to help exceptional children achieve the greatest possible personal self-efficiency and academic success.**

1.2. BASIC TERMS AND /OR CONCEPTS IN SPECIAL NEEDS EDUCATION

- **1. Special needs education:**
- Refers to specially designed instruction to meet the unique needs of exceptional children.
- It exists to help exceptional children in **developing their full human potential.**
- Special needs education is ***both an intervention and instruction.***
- ***When we say it is an intervention, special*** needs education is a purposeful intervention designed to **prevent, minimize and if possible eliminate the obstacles that might keep individuals with disabilities from learning and from full and active participation in school and society.**

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- When we say it is **an instruction, special needs education** requires especially trained professionals (who of its teaching), especially **designed and/or modified curriculum;**
 - ❑ what of its teaching?
 - ❑ specialized or adapted materials (how of its teaching?) and
 - ❑ special classes (where of its teaching?).

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- In short, special needs education aims at making the education system inclusive by educating teachers about **assessing and identifying children with special needs, finding ways to facilitate active learning for all children, and establishing support system.**

2. Impairment

❑ Refers to **any loss/abnormality of;**

- Psychological,
- Physiological or Anatomical
- Structure/function.
- Abnormality of body structure,
- Appearance/organ loss.

Examples: Hearing loss, mental illness, near sightedness, loss of a leg, etc.

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- **Points to consider in impairment:**
- Impairment can be **caused by a genetic abnormality, disease and an accident.**
- Although pathology has involved originally, impairment does not necessarily indicate that a disease is always present.
- **Impairment may or may not lead to a disability.**

3. Disability

- Refers to **any restriction or lack of ability (resulting from impairment) to perform an activity in the manner or within the range considered normal for a person of the same age, culture, and education.**
- It refers to a situation in which a person's **functioning is reduced as a result of significant physical, learning or social problems.**
- **Examples: Inability to read, see, put on ones clothes, hear, etc...**

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- **N.B: - One may have an impairment which results in disability.**
- **However, other popular disabling factors include;**
- **Architectural (physical),**
- **Legal,**
- **Policy,**
- **Social and**
- **Political barriers in the environment.**

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- Various disabling factors reinforce societal expectations that prohibit a person with disabilities from performing activities of which the individual is capable. These are:
 - ❑ **Misconceptions (especially underestimation)**
 - ❑ **Refusal to accommodate to an impairment skill level**
 - ❑ **Unwillingness to allow the person to continue to do those tasks which he still can do.**
 - ❑ **Lack of job opportunities and unavailable social support system.**

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- In addition, existing social value and institutional arrangements such as;
- Inaccessible public building,
- Lack of wide door,
- Lack of accessible public transportation,
- Negative public attitudes towards persons with disabilities, etc.
- In general, the two terms (impairment and disability) are not interchangeable. Impairment may or may not result in disability (e.G. A blind man who became a lawyer is no more a person with disability)

4. Early intervention

- an attempt to prevent or minimize the
- **physical,**
- **cognitive,**
- **emotional and resources limitations** of young children with **biological or environmental risk factors.**
- The intervention program has three basic parts. These are:

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- A) **Preventive:** it refers to measures taken a head of time to avoid factors that may lead to disability or aims to reduce the severity of the condition. This can be achieved through **education and awareness training, nutrition, personal and environmental hygiene and medication.**
- B) **Remedial/Remediation:** it refers to the condition when the impaired part of the body continues to perform its usual activity as a result of the special training/support given to the individual with impairment. **Examples, in schools** a student can be taught academic skills (such as reading, writing, computing or math's skills) social skills (such as getting along with others; following instructions, schedules and other daily routines) personal skills (such as eating, dressing, using the toilet without assistance) and vocational skills (such as career and job skills).
- C) **Compensation/compensatory:** it involves teaching specific skills or the use of devices that enable successful functioning. For example, **wheelchairs for persons who lost their legs and cannot walk; Cochlea Implant for the deaf** and people who have lost their voices compensate their problems by writing what they want to say to others.

5. Rehabilitation

- refers to a process aimed at enabling persons with disabilities to reach and maintain optimal
- Physical,
- Sensory,
- Intellectual,
- Psychiatric, and/or social functional levels
- Thus providing them with the tools to change their lives towards a higher level of independence.

6. Segregation

- refers to separating persons with disabilities from society.
- **Example, teaching children in boarding schools** or placing them in an institution due to their disabilities

7. Mainstreaming

- refers to the return of **handicapped children to the regular classroom from the segregated setting.**
- It is a maximum integration of handicapped students in to the regular classroom.

8. Children with special needs

- all children with impairment, disabilities and handicap;
- street children;
- children with HIV/AIDS and other chronic health problems;
- linguistically,
- culturally and economically minorized/deprived children;
- children who are gifted and talented, etc.

9.Assessment

- is the process of determining whether a child exhibits developmental problem;
- What the problem is?,
- Its cause?,
- Its potential course, its developmental consequence and
- The best approaches to intervention.

10. Identification

- **Detecting the existence of certain impairment or disorder.**
- In the process of identification many individuals are involved such as **parents particularly mother, teachers, and other specialists.**

11. Inclusive education

- refers to an education system that is **open to all learners, regardless of poverty, ethnic backgrounds, language, learning difficulties and impairments.**

12. Inclusive schools

- It means **ordinary (regular schools) open to all children and students regardless of poverty, gender, language, impairment, etc.**

1.3. ORIGIN AND HISTORICAL DEVELOPMENT OF SPECIAL NEEDS EDUCATION (SNE)

- The development of special needs education is **closely related to the society's economic, political, cultural and religious development and civilization.**
- For this reason, the historical development of special needs education is different from country to country.

1.3.1.ORIGINS OF SPECIAL NEEDS EDUCATION

- special needs education actually began over 200 years ago with an amazing story in southern France.
- In **1799**, farmers in southern France found a **young boy in the woods and brought that “wild child” to a doctor in Paris.**
- The child was named **Victor.**
- **Jean-Marc-Gaspard Itard**, the doctor who now is recognized as the *“father of special education,”*

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- In the early **1800s**, **Edouard Seguin**, one of the **Itard's students**, came to the **United States** and began efforts in **this country to educate students with disabilities**.
- In fact, these early efforts were taking root across Europe as well.
- For example, in **Italy**, **Maria Montessori** worked first with children with cognitive disabilities and showed that children could learn at young ages through **concrete experiences offered in environments rich in manipulative materials**.
- Meanwhile, **Thomas Hopkins Gallaudet** began to develop **deaf education**, and **Samuel Gridley Howe** founded the **New England Asylum for the Blind** (later the Perkins Institute).
- **Elisabeth Farrell** initiated **public school classes** for students with disabilities in 1898.

1.3.2. HISTORICAL DEVELOPMENT OF SNE: AN INTERNATIONAL PERSPECTIVE

1. Early History (Before 1800):

- primarily one of misunderstanding and superstition.
- Infants born with disabilities were abandoned into a forest.
- Individuals who were considerably “**different**” from the normal in appearance or behavior were seen as possessed by **demons or evil spirits**.
- people with disabilities were seriously viewed as fools.
- Until the 16thC the general picture of **disability was very dark**.
- Generally, during this time people with disability were **not accepted as totally humans**
- They were **misunderstood, mistreated, and in many cases put to death**.

2. Era of Institutions (1800 – 1900)

- **Institutions for people with disabilities were established in Europe and United States.**
- Considerable support for institutionalization seems to have come from the fact that such a practice;
- **kept these undesirable or physically unattractive persons out of the public eye.**
- The last phase of the 19th century (1898) new ideas began to emerge by scholars.
- **Alexander Graham Bell proposed the ideas of special classes to teach individuals with visual impairment, hearing impairment and mental retardation**

3. Era of Public School /Special Classes (1900 - 1960/70)

- **During this era people with disabilities were provided** by educational services in special classes.
- However, towards the end of this period, **as a result of research findings and questions forwarded from legal point of view**, using special classes as the last option of educational modalities **was seen as a form of segregation.**
- When people hear the term “special classes” they began to develop and manifest a negative attitude towards the term.
- This condition paved the way for the emergence of the following new era in the history of special needs.

4. Era of Accelerated Growth (1960s/70s onwards):

- **At the end of the 20th century, services that had been**
- ❖ **The acceptance of persons with special needs as a human being**
- ❖ **The development of positive thinking and attitude among education professionals and the general public towards persons with special needs**
- ❖ **Measures taken by the courts, professionals, parents, governments, etc concerning the education of children with disabilities enabled these children to have access to take part in continuous and appropriate education with their natural peers. Examples:**
- ❖ **The US Congress passed Public Law 94-142 on “Education for handicapped Act of 1975”**
- ❖ **This Act eventually led to the development of “Individual with Disabilities Education Act (IDEA) of the 1990”.**
- ❖ **The above indicated movements and the initiative taken further by the American Congress enabled to ensure the rights of children with special needs to have access to free and appropriate regular education.**
- ❖ **In addition, legal supports had been practiced to ensure the rights of children to be integrated in the society and have equal access to the resources of their country, take part in any social and economic activity without being restricted by their disability.**

1.3.3. HISTORICAL DEVELOPMENT OF SPECIAL NEEDS EDUCATION IN ETHIOPIA

- ❖ In Ethiopia, religious organizations played a vital role in establishing special schools and centers
- ❖ The first modern special school was established in **1917 E.C by Persian church followers for people with visual impairment at DembiDollo Town.**
- ❖ The second school for people with visual impairment was established in **1936 E.C around kazanchiz but later transferred to Sebeta** where Emperor Haile Sellassie established a building for the blind as a boarding school in 1950 E.C. This school is still functional.
- ❖ **In 1951 E.C. special school for people with visual impairment** was established in Addis Abba around Kechenie area by the followers of MekaneEyesus Church believers which were later transferred to Baco.
- ❖ In **1956 E.C. Mekanisa School for people with hearing impairment** was established by the Meserete Kirstose Church believers.
- ❖ **In 1959 E.C. the Alpha Special School for the deaf** was established around Bole area by the American Missionaries. Now it is under the ministry of education and gives service as a special day school for the deaf.
- ❖ In **1980 E.C. special school for people with mental retardation** was established by **MekaneEyesus around kazanchiz area, Addis Ababa.**
- ❖ From **1986 E.C. on wards (after the implementation of the new Educational and Training Policy)** so many special classes/units within regular schools were established for people with visual impairment, for people with hearing impairment and for people with mental retardation.

1.4. INTERNATIONAL AND NATIONAL LEGAL AND POLICY DOCUMENTS ON SNE

1.4.1. INTERNATIONAL LEGAL AND POLICY DOCUMENTS ON SNE

1. Convention on the Rights of Children (CRC)

- According to this convention article one, a child means every human being below the age of 18 year. From this convention:
- States recognize that children with disabilities should **enjoy a full and decent life.**
- States recognize the right of the a child with a disability **to give special care**
- States recognize the special needs of a child with a disability should ensure **effective access to education,**
- **healthcare, social security, rehabilitation services, preparation for employment** etc.

2. World Declaration on Education for All (EFA)

- The education for all (EFA) movement is concerned with **ensuring access to basic education for all.**
- It was launched at Jomtein, Thailand in 1990. The Declaration materializes the human right to education pronounced in the Universal Declaration of Human Right in 1948 Article 26.

3. United Nations Standard Rules on the Equalization of Opportunities for PWDs

- The purpose of the rules is to ensure that girls, boys, women and men with disabilities as a member of their societies may exercise the same rights and obligations as others.**
- States should take appropriate actions to remove the existing obstacles that prevent people with disabilities from exercising their rights and freedoms, and from participating fully in the activities of their societies.**

4. Salamanca Framework for Action

- The government of Spain in co-operation with UNESCO and held in Salamanca in June 1994.
- The guiding principle of the framework is that:
- ***“Schools should accommodate all children regardless of their physical, intellectual, emotional, linguistic or other conditions. It is the schools to find ways of successfully educating all children, including those who have serious disadvantages and disabilities”.***

1.4.2. NATIONAL LEGAL AND POLICY DOCUMENTS ON SNE

Constitution of the Federal Democratic Republic of Ethiopia-1995: Article 41(5) of the

- ✓ Constitution sets out the State's responsibility for the provision of necessary rehabilitation and support services for people with disabilities.

2. *Labor Proclamation, No. 377/2003, amended by Labor Proclamation No. 494/2006: It states that it*

- ✓ is unlawful for an employer to discriminate against workers on the basis of nationality, sex, religion, political outlook or on any other conditions.

3. The Federal Civil Servant Proclamation- No. 515/2007:It offers for special preference in the recruitment,

- ✓ promotion, and deployment, among others, of qualified candidates with disabilities. This provision is applicable to government offices only.

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Proclamation concerning the Rights to Employment for Persons with Disabilities-No. 568/2008:The

- **proclamation makes null and void any law, practice, custom, attitude and other discriminatory situations that limit equal opportunities for persons with disabilities.** It also requires employers to provide appropriate working and training conditions; take all reasonable accommodation measures and affirm active actions, particularly when employing women with disabilities; and assign an assistant to enable a person with disability to perform their work or follow training.
- 5. Building Proclamation- No. 624/2009: provides for accessibility in the design and construction of**
 - any building to ensure suitability for physically impaired persons.

1.5. CATEGORIES OF PERSONS WITH SPECIAL NEEDS

- **Sensory and motor Impairments and/or Disability**
 - ✓ People with visual impairment
 - ✓ People with hearing impairment
 - ✓ People with physical and health impairment
 - ✓ People with communication disorders
- **Cognitive Impairments and/ or Disability**
 - ✓ People with emotional and behavioral disorder
 - ✓ People with intellectual disability
 - ✓ People with learning disabilities
- **Gifted and Talented Children and Other Vulnerable groups**
 - ✓ Gifted and/or talented individuals,
 - ✓ Individuals with different vulnerability groups,etc

1.7.INCLUSION AND INCLUSIVE EDUCATION

1.7.1. CONCEPTS OF INCLUSION AND INCLUSIVE EDUCATION

- ☐ Inclusion in education/service refers to —an ongoing process aimed at offering quality education/services for all.
- ☐ It is respecting diversity and the different needs and abilities, characteristics and learning expectations of the students and communities
- ☐ It is the elimination of all forms of discrimination.

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- ❑ Inclusive services at any level are quality provisions **without discrimination or partiality** and **meeting the diverse needs of people**.
- ❑ It is seen as a **process of addressing and responding to the diversity of needs of all persons** through increasing participation in learning, employment, services, cultures and communities, and reducing exclusion at all social contexts.
- ❑ It involves **changes and modifications in content, approaches, structures and strategies**, with a common vision which covers all people, a conviction that it is the responsibility of the social system to educate all children, employ and provide social services.

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- ❑ **Inclusion** is defined as having a wide range of strategies, activities and processes that seek to **make a reality of the universal right to quality, relevant and appropriate education and services.**
- ❑ It seeks to enable communities, systems and structures in all cultures and contexts to **combat discrimination, celebrate diversity, promote participation and overcome barriers to learning and participation for all people.**

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- ❑ It is part of a wider **strategy promoting inclusive development.**
- ❑ The goal of creating a world where there is **peace, tolerance, and sustainable use of resources, social justice,** and where the basic needs and rights of all are met.

- **Inclusive education is the process of systematically bringing together all children with or without disabilities regardless of the nature and severity of disability in natural environment where children learn and play.**
- It is the practice of educating students with special needs in regular schools instead of in special education classes.
- Its main tenet is that **all children with disabilities have a right to be included in naturally occurring settings and activities with their neighborhoods, siblings and friends.**

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- Globally, inclusive education is understood as a process towards creating a system of education that meets the needs of all children, recognizing that many different groups are currently excluded.
- Inclusive education happens when children with and without disabilities learn and play together in the same class.

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- The curriculum in inclusive education requires educators to develop appropriate curriculum and employ effective instruction to benefit all learners.
- Inclusive curriculum is a school curriculum that emphasizes diversity and accommodates the diverse needs of all children in the class room.
- The inclusive curriculum expands inclusive education by including children with varying abilities.

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- Inclusive education gives high consideration for **curriculum, teaching approach and learning environment.**
- Inclusive curriculum should be appropriate to the child's cultural, social, environmental context and relevant to the child's unique characteristics such as interests, abilities, and learning needs.
- Inclusive curriculum demands a broad common goal defined for all learners.

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- The curriculum should have flexibility for responding the cultural, religious and linguistic diversity of all learners.
- Generally, inclusive curriculum needs curriculum flexibility.

Adapted and modified contents: in inclusive education, making appropriate adaption or modification to the

- ✓ contents of the curriculum is essential. Inclusive curricula are constructed flexibly to allow adaptation and modification not only at school level but also to meet individual students; needs and teachers' style of working.
- ✓ Under these circumstances, every student is likely to receive a curriculum experience which is **tailored to his/her needs but within the context of a common framework or national educational objectives.**
- ✓ Under these circumstances, the curriculum does not require students to be taught separately from their peers in order to follow an individual program.

b. Adapted teaching method: the teaching methods should promote inclusive learning environments. Teachers

- ✓ to be aware of the diverse educational needs of students with disabilities. They need to acknowledge **the learners preferences and learning styles.**
- ✓ This may require the adjustment of teaching methods, materials, and settings, rather than adjusting the children to existing methods.
- ✓ More specifically, teaching strategies that **give children a choice of multisensory materials and activities at different levels are found to be useful.**

C) Adapted assessment: Inclusive curriculum uses assessment to monitor individual progress.

- ❖ **Assessment is not to decide which learners should be promoted to next grade or to rank learners by setting a set of norms.**
- ❖ When assessment is **used for selection or ranking**, it is inevitable that learners with special educational needs will score least or lags behind. Thus, this **may stigmatize them as “failures” and “de-motivate them”**.
- ❖ Assessment in inclusive curriculum is mainly for **the purpose of promoting learning and guiding teaching**.
- ❖ This means, assessment is conducted to **know what learners know, what learners can do or has experienced**. In an inclusive classroom, assessment should meet the following criteria.
- ❖ When learners have not mastered particular tasks, teachers should diagnose why this occurred and this gives them opportunities to reevaluate their teaching method and content. This is what is referred to as the formative assessment.
- ❖ It should **provide feedback to learners and their parents about the progress of their children**.
- ❖ It should **focus on identifying what students achieved (mastered) and what is not achieved (i.e., criterion referenced assessment) rather than putting them in some sort of order of merit (i.e., norm referenced assessment)**
- ❖ Method of assessment of **learners with special needs should take account of their specific disabilities**. For instance, a blind learner may need to be tested in Braille or orally, a deaf learner may need to be tested using sign language, a learner with reading disability may need more time during examination.

II) Inclusive teaching

- ☐ **takes into account the potentiality and learning style of all learners. Individual abilities, interest and learning styles should be scrutinized in advance and diversity is not regarded as problems to be solved, rather diversity is acknowledged and respected.**
- ☐ **Inclusive teaching practices support the learning of all students.**
- ☐ **Students in inclusive school should be taught in accordance with their learning preferences and learning style.**
- ☐ **The learning approach in inclusive teaching is child-centered.**
- ☐ **Individual children learn and develop in different ways, at different rates and**
- ☐ **they seek learning environment which respond to their special educational needs of each child.**

III) The learning environment in inclusive school supposed to be suitable for all learners with or without disabilities.

- ✓ Creating a welcoming and accessible environment in which children can learn is a major tenet of inclusive education. These environments for learning include:
- ✓ **Classroom Situation (both the physical and social environments):** - the learning environment in inclusive schools has to be accessible to all learners and it should be free from physical and social barriers.
- ✓ **Manageable class size and teaching load:** - the learning environment in inclusive classroom should comprise manageable class size.
- ✓ Large class sizes can be considered as a barrier to the inclusion of children with disabilities in all countries especially in developed countries.
- ✓ In addition to class size, teaching hours and number of children with special needs in a classroom should be considered

1.7.2. MODELS OF INCLUSION

1) Medical or Individual-Based Model

- ❑ In this model, the unit of analysis is the person.
- ❑ The cause of individual deficits or failure rests within the individual and his or her physical body.
- ❑ Individuals do not progress satisfactorily because of inadequate cognitive, behavioral, sensory, motor, medical and physical characteristics.
- ❑ In the same way a disease is understood as being “owned” by the infected individual and hence disability is considered as the exclusive property of the student.
- ❑ Intervention strategies under the **individual-deficit orientation** involve assessing individual attributes and include such strategies as correcting the problem through medication and nutrition.
- ❑ It is unfortunate that this paradigm remains linear and mechanistic and does not account for the complex experiences of individuals in social settings.
- ❑ Disability is an individual problem. Thus, they should be —treated —changed "or —improved" and made more —normal.
- ❑ It involves the philosophy of “**change the child to fit the system**”.

2) Environment-Based or Social Model

- In this model, the unit of **analysis is primarily the environment**.
- Since behavior is learned individuals fail to progress because of inappropriate or inadequate environmental circumstances in which they develop or learn.
- Within the **environment-deficit orientation**, typical intervention strategies include evaluation of the learning environment, matching characteristics of teachers and related service providers to individual student characteristics, evaluating student teacher-ratios, and facilitating family involvement.
- This model is criticized for using a mechanistic approach to human complex behavior, undermining the role of the individual by promoting single-cause-effect understanding on human behavior.
- It involves the philosophy of “**change the system to fit the child**”.

3) System-Based Model

- This model holds that developmental processes do not occur in a psychological vacuum, but rather that individual child development is influenced by factors in the immediate environment as well as society and culture as a whole.
- Rather than focusing on the singular like emotion, cognition, or learning, tend to attempt to understand developmental change in its entirety, that is, the whole child and the whole environment .
- The result of the individual factors or the environmental factors but the interplay of the two factors.
- It involves the philosophy of “ change the child and the system to fit the child and system.”

2The Rational of Moving Towards Inclusion

- ❑ Among others, the following are the reasons for paradigm shift of moving towards inclusive education system as a whole:
- ❑ Inclusive education enhances the attainment of the **goal of education for all**. Education cannot be for all until it is received by all.
- ❑ Inclusion promotes a sense of **cooperation and the feeling of togetherness** in the learner
- ❑ The population of students with special needs is becoming increasingly diverse. As diversity increases, so does the diversity of the students that compose the schools.
- ❑ To meet this **diverse student population needs**, the schools should respond with varied services for all its students
- ❑ Inclusive education is **relatively cost effective**; it accommodates all learners in the same environment using the same facilities.
- ❑ The inclusion movements promises, through innovative teaching techniques, research and the collaboration of many experts, to provide caring learning experiences in which all **students have an opportunity to be successful**.

1.7.3. Benefits of Inclusion

A) Benefits for Students with Special Needs Education

- Appropriate models of behavior.
- Increased social initiations, interactions, relationships and networks
- Increased achievement of individualized educational program (IEP) goals
- Greater access to general curriculum
- Enhanced skill acquisition and generalization in their learning
- Attending inclusive schools increases the probability that students with SEN will continue to participate in a variety of integrated settings
- Improved school staff collaboration to meet these students' needs and ability differences
- Increased parental participation to meet these students' needs and ability differences.

B/ Benefits for persons without Special Needs Education

- opportunities for interacting with their age peers
- serve as peer tutors during instructional activities
- Play the role of a special buddy' during lunch, in the bus or playground.
- Gain knowledge of a good deal about tolerance, individual difference, and human exceptionality.
- Learn that students with SEN have many positive characteristics and abilities.
- Have chance to learn about many of the human service profession such as special education, speech therapy, physical therapy, recreation therapy, and vocational rehabilitation.
- Have increased appreciation, acceptance and respect of individual differences.
- Get greater opportunities to master activities by practicing and teaching others
- Have increased academic outcomes
- have opportunity to learn to communicate, and deal effectively with a wide range of individuals

C/Benefits of inclusion for Teachers

- learn new ways to teach different kinds of students.
- They develop more positive attitudes and approaches.
- They have greater opportunities to explore new ideas.
- They can encourage their students to be more interested, more creative and more attentive
- They can experience greater job satisfaction and a higher sense of accomplishment when all children are succeeding in school to the best of their abilities.
- They get opportunities to exchange information about instructional activities and teaching strategies.
- They benefit from Developing teamwork and collaborative problem-solving skills to creatively address challenges regarding student learning

D/Benefits for Parents/Family

- Learn more about how their children are being educated in schools with their peers in an inclusive environment
- Become personally involved and feel a greater sense of accomplishment in helping their children to learn.
- Feel valued and consider themselves as equal partners in providing quality learning opportunities for children.
- Learn how to deal better with their children at home by using techniques that the teachers use in school.
- Find out ways to interact with others in the community.
- Know that their children—and ALL children—are receiving a quality education.
- Experience positive attitude about themselves and their children by seeing their children.

E/Benefits for Society

- Mainstream schools bring in the students into local communities and neighborhoods.
- Helps break down barriers and prejudice that prevail in the society towards PwD.
- Communities become more accepting of difference, and everyone benefits from a friendlier, open environment.
- Meaningful participation in the economic, social, political and cultural life of communities own cost effective non-segregated schooling system that services both students with and without special needs education.

1.7.5.Ultimate Goal of Inclusion

- The goal of inclusion is to create schools where everyone belongs.
- By creating inclusive society, we ensure that there's a welcoming place in the community for everyone after their school year's end.
- Students educated together have a greater understanding of difference and diversity.
- Students educated together have fewer fears about difference and disability.
- An IS culture creates better long-term outcomes for all students.

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- Inclusive society is a necessary precondition for inclusive growth is a society which does not exclude or discriminate against its citizens on the basis of disability, caste, race, gender, family or community, a society which levels the playing field for investment' and leaves no one behind.
- Inclusive growth, which is equitable, that offers equality of opportunity to all as well as protection in market and employment transitions results from inclusive society.

1.7.6. Features of Inclusive Environment

- members feel respected by and connected to one another.
- welcomes all people, regardless of their disability and other vulnerabilities.
- It recognizes and uses their skills and strengthens their abilities.
- respectful, supportive, and equalizing. An inclusive environment reaches out to and includes individuals with disabilities and vulnerabilities at all levels
- It has the following major characteristics:
 - ☐ it ensures the respect and dignity of individuals with disabilities
 - ☐ it meets current accessibility standards to the greatest extent possible to all people with special needs
 - ☐ provides accommodations willingly and proactively
 - ☐ Persons with disabilities are welcomed and are valued for their contributions as individuals.

Inclusive Environments

- It develops whole-school/environment processes
- It recognizes and responds to the diverse needs of their individuals and ensuring quality provisions for all through appropriate accommodations, organizational arrangements, resource use and partnerships with their community.
- It is also deeply committed to the belief that all persons can learn, work and be productive.
- It involves restructuring environment, culture, policy, and practice.
- It promoting pro-social activities
- It makes provides services and facilities equally accessible to all people
- It involves mobilizing resources within the community
- It is alert to and uses a range of multi-skilled personnel
- strong links with, clinicians, caregivers, and staff in local schools, work place, disability services providers and relevant

2.8. Barriers to Inclusion

- societal values and beliefs- particularly the community and policy makers negative attitude towards students with disability and vulnerabilities.
- Economic factors- this is mainly related with poverty of family, community and society at large
- Lack of taking measures to ensure conformity of implementation of inclusion practice with policies
- Lack of stakeholders taking responsibility in their cooperation as well as collaboration for inclusion
- Conservative traditions among the community members about inclusion
- Lack of knowledge and skills among teachers regarding inclusive education
- Rigid curricula, teaching method and examination systems.
- Fragile democratic institutions that could not promote inclusion
- Inadequate resources and inaccessibility of social and physical environments
- Large class sizes
- Globalization and free market policy that make students engage in fierce completion, individualism.
- Using inclusive models that may be imported from other countries

UNIT TWO: SENSORY, MOTOR, HEALTH AND COMMUNICATION IMPAIRMENTS

2.1. Children with visual impairment

2.1.1. Definitions of visual impairments:

- ☐ **VI can be defined from different points of view differently.**
- ☐ It can be defined from medical (clinical) or legal and educational perspectives.
- ☐ The definitions given to visual impairment are of two types. Such as:
 1. The legal definitions focus on the measure of visual acuity and field of vision
 2. Educational definitions is whether there exists a residual vision that enables an individual to learn using his/her sense of sight or not.

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- This sensory impairment can be classified into two in terms of severity :
- partially sighted and
- blind.
- **Partially sighted/low vision** learners are those having significant visual problems but still use their vision as their primary sense for learning and
- **Blind** learners are those whose visual impairment is so severe that they must rely on senses other than vision to function adequately.

Cont...

- Measurement units typically are expressed in reference to Snellen chart notations of visual acuity.
- This chart consists of symbol (number or letters) that decrease in size; it is read at a distance of 20 feet.
- Symbol size corresponds to the standard distance at which the people with normal vision can recognize the symbol.
- Acuity of 20/20 means that a person can read the symbol at a distance of 20 feet, a ratio considered representative of normal vision.

Cont...

- In other words, an object that a person with a normal vision (20/20) can see at 200 feet must be brought to 20 feet before the person with 20/200 acuity can read it.
- **Blind** : visual acuity of 20/200 or less in the better eye, with correction/ or whose visual field of less than 20degree or less in the better eye.
- **Partial sight individuals** are with visual acuity ranging between 20/70 – 20/200
- When vision is more significantly impaired (20/400 or worse) or when very young or handicapped persons are tested, it is frequently necessary to test vision at distances of less than 20 feet.
- In these cases, the numerator is less than 20, usually 10 or 5. A reading of 10/200 means that an object distinguished by persons with normal activity at 200 feet must be brought as close as 10 feet for recognition.

Cont...

- As the denominator in the measure of visual acuity increases, visual acuity decreases while the severity of visual problem increases.

Cont...

- **Field of vision-** refers to the angular distance or space, which is covered by vision.
- The **normal field of vision is 180°**. The legal definition of visual impairment relies heavily on the measurement of visual acuity and range/field of vision.
- Legally, it is defined blindness as having central visual acuity of 20/200 or less in their better eye with correction, or having the peripheral vision contracted to an extent in which the widest diameter of the visual field covers an angular distance not longer than 20 degrees.

Cont...

- The term **blindness educationally** typically refers to vision loss that is not correctable with eyeglasses or contact lenses.
- Blindness may not mean a total absence of sight.
- However, some people who are considered blind may be able to perceive slowly moving lights or colors.
- The term **low vision educationally** is used for moderately impaired vision. People with low vision may have a visual impairment that affects only central vision the area directly in front of the eyes or peripheral vision the area to either side of and slightly behind the eyes. It can be corrected with corrective lens/enlarged print.

2.1. 2 Major causes of Visual Impairments

- **1.Refractive errors:**
- **Myopia or nearsightedness:** is caused by increased curvature of the eyeball (where the eyeball is longer than normal from front to back) resulting The refracted image comes into focus **in front of the retina instead of on it**. Nearsighted glasses, contact (**concave**) lenses, and laser refractive surgery are all used to move the image from in front of the retina to on the retina itself!
- **Hyperopia or farsightedness:** is caused by a decreased curvature of the eyeball (the eyeball being shorter than normal from front to back) resulting in light rays focusing beyond the retina, causing difficulty in seeing objects that are nearer. Farsighted glasses, **contact (convex) lenses, and refractive surgery are all used to move the image from behind the retina on to the retina itself!**
- **Astigmatism:** Is the outcome of **irregularities or unevenness in the cornea**, causing light to be refracted differentially rather than at one point where part of the image is correctly focused, resulting in **blurred or distorted vision**.
- **Glaucoma:** Glaucoma is a condition that is related to **a significant increase in the pressure exerted by the fluid (aqueous humor) within the eye itself**. Usually the fluid is secreted and circulated in the eye. If the pressure of this fluid becomes too high, **internal damage (damage of the optic nerve)** which can result in **severe loss of sight or tunnel vision**. The person sees only the central of the visual field. This **can lead to permanent loss of vision**. But, if it has been detected early enough, glaucoma can be treated by controlling the pressure in the eye.
- **Cataract:** is a condition where the lens becomes **Opaque (cloudy)** instead of remaining transparent. This results in a loss of visual acuity with light perception being adversely affected. This can be surgically corrected.

2. Muscle Disorder or Defects

- Disorders relating to ciliary muscles around the eye are:
- **Nystagmus:** A circular or side-to-side, rapid involuntary movement of the eyeball.
- **Strabismus:** An inability of the muscles to pull equally, resulting in crossed eye (internal strabismus) or eyes pulled outward (external strabismus). Surgery can correct many of these conditions.
- **Amblyopia:** A loss of vision due to muscle imbalance that results in double vision. Here the brain attempts to reduce the confusion thus created by repressing the vision in one eye and the unused eye may therefore atrophy, resulting in loss of sight. Here the child can put a patch on the unaffected eye to force the affected eye to focus. Surgery also helps.

3. Diseases

- the potential causes of visual impairment. The causal factors may be of viral, bacterial, or enzymatic abnormalities. Some might be treated; while others cause permanent and complete damages on our vision. It can include the following:
 - **Diabetic Retinopathy:** A disease of the small blood vessels that nourish the retina is the most common eye complication of diabetes mellitus, a disease in which glucose or sugar is not properly used by the body, along high levels of sugar to build up in the blood and urine. It is impaired vision due to hemorrhage (great flow of blood) and the growth of new blood vessels in the area of the cornea.
 - **Sickle-cell retinopathy**
 - Retinopathy of prematurity (retrolental fibroplasias) in its advanced stages causes an abnormal fibrous condition in the eye of a premature infant. Sickle-cell retinopathy involves blockages and hemorrhages of retinal blood vessels that occur in association with sickle-cell anemia.
 - **Infectious diseases:** Trachoma, syphilis, measles, small pox can cause visual impairment.
4. **Accidents and lack of nutrition (vitamin A):** Exposure for very powerful light, being hit by bullet or any other thing, acid attack,... are among the accidents we may encounter and face visual impairment.

2.1.3 Symptoms and Identification of VIS:

- **Rubbing eyes:** - The rubbing may be observed in excessive amounts or during close visual work.
- **Shutting or covering one eye:** - a student who is having difficulty seeing may close one eye, tilts, or thrust the head forward.
- **Light sensitivity:** - Some students with an undetected vision problem may demonstrate unusual sensitivity to bright or even normal light.
- **Difficulty with reading:** - Unusual amount of difficulty in reading or other work requiring close use of the eyes.
- **Losing place during reading:** - A tendency to lose his/her place in a sentence or page while reading. 14
- **Unusual facial expressions and behaviors:** - Unusual amount of squinting, blinking, frowning, or facial distortion while reading or doing things.
- **Achievement disparity:** - disparity between expected and actual achievement is one of the indications of visual problem.

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- **Eye discomfort:** - the student who complains of burning, itching, or scratchiness of the eyes may be experiencing a vision problem.
- **Holding reading materials at an inappropriate distance** holding reading materials too close or too far or frequently changing the distance from near.
- **Discomfort following close visual work:** - the student who complains of pains or aches in the eye, Headaches, dizziness, or nausea following close visual work may need corrective glasses.
- **Difficulty in distance vision and Blurred or double vision**
- **Reversals:** - a tendency to reverse letters, syllables, or words may be an indication of impaired vision
- **Letter confusion:** - a student who confuses letters of similar shape (o and a, c and e, similarly n, and m, etc) may have impaired vision.
- **Poor spacing:** - poor spacing in writing and difficulty in “staying on the line” may be an indication of vision problem.

2.1.4 Support Systems of Children with Visual Impairments

- **Educational support for children with VI: Educators believe that the most important visual consideration is functional visual efficiency, or how well children use their vision, rather than the particular measure of visual acuity.**
- **Teachers of children with visual impairment are often taught of in conjunction with specialized equipment and materials such as Braille, canes, tape recorders and magnifying devices.**
- **Whatever the range of visual deficit, it is crucial that any remaining vision is utilized to a greater extent possible.**
- **The child needs to be encouraged to optimally utilize his/her tactile sense or auditory modality similar to that of hearing children.**
- **Both the parents and the schools make their own contribution in intervening to the difficulties of children with visual impairment. Yesseldyk and Algozine (1995) have stated ten tips for teachers of students with visual impairments.**

Cont...

- ☐ Reduce distance between student and speaker as much as possible.
- ☐ Reduce visual distracting as much as possible.
- ☐ Reduce disorder on classroom floor and produce free access to door and key classroom spaces.
- ☐ Seat students near chalkboard or overhead projections, or give them the freedom to move close areas of instruction.
- ☐ Avoid partially opening cabinets, storage areas, and classroom doors; ascertain that fully opened or closed doors as sager.

Cont....

- ☐ Use auditory cues when referring to objects in the classroom and during instructional presentations,
- ☐ When presenting visually dependent material, verbalize written information, describe pictures, and narrate non-verbal sequences in videotapes or movies.
- ☐ Use complete sentence to provide additional content.
- ☐ Reduce unnecessary noise it helps focus cement of instructional presentations.
- ☐ Keep instruction materials in the same place so students can find them easily,
- ☐ Make sure glasses and other visual aids are functioning property.

2.2 Hearing Impairments

2.2.1 Definition of Hearing Impairments (HI):

- Passonella and Care (1998), defined hearing impairment as a generic term indicating a **continuum of hearing loss from mild to profound**, which include sub classifications of **hard of hearing** and **deaf**.
- Any kind and level of obstruction faced by our sense of hearing is collectively named hearing impairment.

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- In Ethiopia, there is no document which helps to understand both the terms hard of hearing and deafness.
- Instead, it is described with a single term consisting negative notions such as “idiots” which is to say those who cannot be educated or do not at all understand.
- Such an erroneous understanding of the hearing impaired is evident by the widely used Amharic term “Donkoro” meaning as described in Amharic Dictionary; an individual whose **hearing organ does not at all function; mentally handicapped; and who lack the ability to understand any language.**

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- In modern world, there are two perspectives or ways of defining hearing impairment: physiological and educational definitions.
- **Physiological definition:**
 - ☐ **Deaf:** is a condition of hearing loss about 90 Decibel (dB) or greater.
 - ☐ **Hard of hearing:** is a condition of hearing impairment of less than 90 dB.
- **Educational definition of hearing impairment:**
 - ☐ **Hard of hearing:** is a term that describes persons with enough residual hearing to use hearing (usually with hearing aid) as a primary modality for the acquisition of language and in communication with others. This condition can affect the child's educational performance to some extent.
 - ☐ **Deafness:** is a term that described people whose sense of hearing cannot process linguistic (verbal) form of information.

2.2.2 Major Causes of Hearing Impairments:

- Hearing impairment is caused by various congenital and acquired conditions of outer, middle and inner ears
- ❑ **congenital conditions of the middle ear Include:**
 - ✓ chromosomal and skeletal abnormalities
 - ✓ Malformation of pinna is a known congenital condition of outer ear that causes hearing impairment.
 - ✓ like rubella (German measles). Cytomegalovirus and bacterial infection such as syphilis., circulatory disorders and infections
- ❑ **Acquired conditions of outer ear.**
 - ✓ Injury of the pinna,
 - ✓ foreign bodies: poking an object into the middle ear
 - ✓ excessive wax,
 - ✓ Noise damage
 - ✓ ineffective conditions like otitis media (inflammation of the middle ear); and facial paralysis)
 - ✓ infection of the outer ear, and
 - ✓ Tumor/cancer
 - ✓ ototoxic drugs, and conditions involving vertigo (Tirussew, 2000).

2.2.3 Symptoms and Identification of HIs:

- There are several **symptomatic behaviors** that are used to determine whether a hearing impairment may be present.
- Once a child is suspected and determined that he/she has a HI,
- Then, parents or teachers must refer the child for further **audio-logical or medical evaluation** to determine the type or degree of hearing loss.
- **These evaluations will help to decide:**
 - on the type of hearing aid needed, and
 - on the nature of school placement.
- In order to understand the symptoms of hearing impairment, it is helpful for parents and teachers to have knowledge of the normal auditory development.

Some behavioral warning signs of HI

- ❖ Inattention, restlessness, distraction of others, more responsiveness in quiet conditions
- ❖ Complaints of earache, popping a visible discharge from the ear
- ❖ Giving inappropriate answers to questions, watching and following what other children do
- ❖ Louder or softer voice than is usual
- ❖ Slowness in responding to simple verbal instruction, with frequent requests for repetition
- ❖ Searching visually to locate a sound source or turning head to give one side an advantage
- ❖ Needing to sit nearer a sound source than is usual or asking for volume on TV, tape or record player to be tuned up.
- ❖ Some irritability or typical aggressive outbursts more frequent behavioral upsets in school
- ❖ Reluctance to participate in oral activity and little interest in following story
- ❖ Failure to turn immediately when called by name unless other visible signals are given
- ❖ Tiring easily, poor motivation, some stress signs such as nail-biting
- ❖ Particularly difficulties in verbally related skills such as reading sound blending and discrimination and writing with better skills in practical areas
- ❖ Speech limited in vocabulary or structure and use of gesture
- ❖ Best work in small group.

- Yesseldyke and Algozzine (1995) also listed **ten potential signs CWHI in the school.**
 - ☐ Difficulties following oral presentations and directions
 - ☐ Watches lips of teachers or other speakers very closely
 - ☐ Turns head and leans toward speaker
 - ☐ Uses limited vocabulary
 - ☐ Uses speech sounds poorly
 - ☐ Shows delayed language development
 - ☐ Does not often respond when called from behind
 - ☐ Generally inattentive during oral presentation
 - ☐ Constantly turns volume up on radio or television
 - ☐ Complains of earaches, has frequent colds or ear infections, has ear discharge.

2.2.4 Support Systems of Children with Hearing Impairments

- Once a hearing problem is detected and diagnosed, intervention should follow.
- **Early intervention** is largely accepted to revert the problem before it develops worse, **both parents and teachers can play a significant role in the intervention process but here teachers (school) interventions are focused.**
- It should be stressed that the appropriate teaching environment **for hard-of hearing should be more or less similar with that of the hearing students.**
- It is always important to remember that **placement in regular schools (less restrictive programs) is a function of the degree of hearing loss.**

Cont...

- According to Heward&Orlansky (1988), the following characteristics are among these considered most critical to the effective teaching of hearing impaired students.
- **Providing language instruction**
 - **Teaching small groups of SWHI, who function on different levels.**
 - **Developing and adopting instructional materials and enhancing positive self-concept**
 - **Using information from various assessment procedures to develop individualized educational program (IEP) and Dealing with crisis calmly an effectively.**

Cont...

- In practical instructional process, the teachers should pay attention for PWHI to the following ten tips (Yesseldyke and Algozzine, 1995).
 - ☐ Reduce distance between student and speaker as much as possible
 - ☐ Speak slowly and stress clear articulation
 - ☐ Reduce background noise as much seat student near center of desk arrangements and away
 - ☐ Use face-to- face contact as much as possible
 - ☐ Use complete sentences to provide additional context during conversation or instructional presentations
 - ☐ Use visual cues when referring to objects in the classroom and during instructional presentations
 - ☐ Have classmates who take notes during oral presentations for student to transcribe after the lessons
 - ☐ Encourage independent activities, co-operative learning and social skills
 - ☐ Be sure that the hearing and the hearing impaired are functioning properly

2.3 Physical and Health Impairments

- 2.3.1 Definition of Physical and Health Impairment
- **physical disability** is a condition that significantly limits one or more basic physical activities in life (i.e. walking, climbing stairs, reaching, carrying, or lifting).
- These limitations hinder the person from performing tasks of daily living.
- Physical disabilities are difficulties associated with sitting, standing, getting into position, moving, communicating, using and manipulating classroom tools and materials and self-care.
- Many but not all, physical disabilities are **orthopedic or neuromotor impairments**.
- The term ***orthopedic impairment involves the skeletal system-bones, joints, limbs, and associated muscles.***
- ***Neuromotor impairment involves the central nervous system, affecting the body's ability to move, use, feel, or control certain parts of the body.***

2.3.2 Classifications of Physical Impairments

- physical Disability can be classified based on two classification systems.
- First, it is classified based on the **degree of severity of the physical disability. This involves:**
 - (A) Mild physical disability:** individuals with mild physical disability can walk without aids and may make normal developmental progress.
 - (B) Moderate physical disability:** individuals with moderate physical disability can walk with braces and crutches and may have difficulty with fine motor skills and speech production.
 - (C) Severe physical disability:** individuals with severe physical disability are wheelchair dependent and may need special help to achieve regular development

Cont...

❖ The second classification system is primarily based on the **affected area**. This includes: **two major types of physical disabilities**:

- 1) the neurological system (which are caused by brain, spinal cord and nerve related problem), and**
- 2) musculo-skeletal system (which are caused by problems in muscles, bones, ligaments and joints due to various causes).**

2.3.3 Causes and Types Physical Disabilities

- **Neurological System related problems**
- some of the physical disabling conditions due to problems in the neurological system are the following.
 - 1) Cerebral Palsy (CP): is a group of movement disorders that results from damage to the brain.**
- **“Cerebral” refers to the brain. “Palsy” refers to paralysis, which accounts for the lack of muscle control associated with this disability**
- **. It is caused by damage to the brain occurring before, during, or shortly after birth.**
- **Causes of brain damage may include, but are not limited to mother’s illness during pregnancy (such as German measles), Rh incompatibility (a bloody conflict between mother and fetus),**

Cont...

- There are **five types of cerebral palsy**, each displaying **different symptoms**, and each determined by where the damage occurred in the brain. It is possible for people to have a combination of these.
- ❑ ***Spastic*** - tense, contracted muscles. The most common type of cerebral palsy
- ❑ ***Athetoid***- constant, uncontrolled motion of arms, legs, head, and eyes
- ❑ ***Ataxic*** - poor sense of balance, often causing falls and stumbles.
- ❑ ***Rigidity*** - tight muscles and inability to move them.
- ❑ ***Tremor*** - ***uncontrollable shaking, interfering with coordination***

2) Spina bifida

- **a condition that affects the spine and is usually apparent at birth.**
- **It is a type of neural tube defect (NTD).**
- **Spina bifida can happen anywhere along the spine if the neural tube does not close all the way.**
- **When the neural tube doesn't close all the way, the backbone that protects the spinal cord doesn't form and close as it should.**
- **Spina bifida happens in the first weeks (21-26 days) of pregnancy, often before a woman knows she's pregnant.**
- **It occurs during the third and fourth weeks of pregnancy when a portion of the fetal spinal cord fails to properly close.**
- **As a result, the child is born with a part of the spinal cord exposed on the back.**
- **Although folic acid is not a guarantee that a woman will have a healthy pregnancy, taking folic acid can help reduce a woman's risk of having a pregnancy affected by spina bifida.**

Types of Spina Bifida:

- (1) **Spina bifida "Occulta"** means "hidden" in Latin is the mildest type of spina bifida. It is sometimes called "**hidden**" **spina bifida**. With it, there is a small gap in the spine, but no opening or sac on the back. The spinal cord and the nerves usually are normal. Many times, spina bifida occulta is not discovered until late childhood or adulthood. This type of spina bifida usually does not cause any disabilities.
- (2) **Meningocele** happens when a sac of spinal fluid (but not the spinal cord) pushes through an opening in the baby's back. Some people have few or no symptoms, while others have problems with their bladder and bowels control. There is usually little or no nerve damage. This type of spina bifida can cause minor disabilities.
- (3) **Myelomeningocele** is the most serious type of spina bifida. With this condition, a sac of fluid comes through an opening in the baby's back. Part of the spinal cord and nerves are in this sac and are damaged. This type of spinal bifida causes moderate to severe disabilities, such as problems affecting how the person goes to the **bathroom, loss of feeling in the person's legs or feet, and not being able to move the legs.**

II. Musculoskeletal Related Problems

- **(A) Muscular Dystrophy (MD)** a group of diseases that make muscles weaker and less flexible over time.
- It is caused by a problem in the genes that control how the body keeps muscles healthy.
- For some people, the disease starts early in childhood. Others don't have any symptoms until they are teenagers or middle-aged adults.
- The most common form of MD in children is Duchene Muscular Dystrophy (DMD).
- The disease almost always affects boys, and symptoms usually begin early in childhood.
- Children with DMD have a hard time standing up, walking, and climbing stairs.
- Many eventually need wheelchairs to get around.
- They can also have heart and lung problems. It is inherited from the mother who has defective gene that regulates dystrophin, which is responsible for the maintenance of muscle fiber.
- The gene is located on the X-sex chromosome, so primarily males will inherit the disorder than females.
- Although physical therapy can be very helpful to reduce the worsening of the problem, unfortunately, at present time there is no known cure or therapy that helps prevent the progression of muscle deterioration.

(B) Myotonic (also called MMD or Steinert's disease).

- The most common form of muscular dystrophy in adults.
- myotonic muscular dystrophy affects both men and women, and it usually appears any time from early childhood to adulthood.
- In rare cases, it appears in newborns (congenital MMD).
- The name refers to a symptom, myotonia -- prolonged spasm or stiffening of muscles after use.
- This symptom is usually worse in cold temperatures.
- The disease causes muscle weakness and also affects the central nervous system, heart, gastrointestinal tract, eyes, and hormone-producing glands.
- Those with myotonic MD have a decreased life expectancy.

- **Health Impairment** is having limited strength, vitality or alertness, including a heightened alertness to environmental stimuli that results in limited alertness with respect to the educational settings.
- This is due to chronic or acute health problems such as asthma, diabetes, epilepsy, a heart condition, hemophilia, lead poisoning, leukemia, nephritis, rheumatic fever, sickle cell anemia, and HIV/AIDS; and adversely affects a child's educational performance.

2.3.4 Causes and Types of Health Impairments

- **1. *Epilepsy***
- a disorder that occurs when the brain cells are not working properly and is often called a seizure disorder
- . It is a neurological condition that involves rapid and unusually brief changes in consciousness accompanied by involuntary movements.
- There's no cure for epilepsy, but the disorder can be managed with medications and other strategies.
- Although the exact cause of a seizure is usually not known, it can result from various conditions, including:
- (1) Developmental brain abnormalities (2) Infections (3) Brain injury/trauma (4) Hypoxia and Hypoglycemia

Cont...

- **Types of Epilepsy:** The following are the three major types of epilepsy.
- **A. Grand mal /Great Illness:** is the most complex seizure as compared to the others. It consists of the following four phases:
- **1. The aura phase:** this phase is characterized by unusual sensory signals, such as strange odors, strange sounds, etc. which are warnings of the impending convulsion.
- **2. The tonic phase:** here there is the actual convulsion which is followed by: (1) Body rigidity (2) Loss of consciousness (3) Falling heavily on the ground (4) Opening of the eyes and Suspending of breathing for about 30 seconds
- **3. The clonic phase:** it is characterized by: (1) Secretion of muscles'' spasms (2) Violent, continuous and jerky movements of limbs and Losing of spinster control
- **4. The coma phase:** in this phase: (1) The individual still remains unconscious (2) The convulsion stops completely and The victim wakes up either immediately or after several hours

- **Petit mal /absence seizure:-**

- **It is characterized by:**

1. Sudden head dropping
2. Frequent blinking of the eyes
3. Dropping/throwing of objects at hands
4. Missing parts of verbal instruction
5. May be loss of consciousness for 5-30 seconds
6. It may occur 50-200 times a day and If left untreated, it may be transformed in to grand mal seizure

Psychomotor seizure (complex petit mal seizure):-

- Here the attacks begin with the subjective experience of sensations, such as peculiar odors, which is accompanied by;
- Psychic disturbances, such as restlessness,
- Unusual fear,
- Aggressiveness,
- Eye blinking,
- Lip smacking,
- Facial grimacing, etc.
- This in turn is followed by:
- Climbing up a table/chair
- (2) undress in public (3) searching things they do not loss and disrobing/laughing or crying for no reason

Diabetes

- Diabetes is a chronic disorder of metabolism.
- It occurs when the body/specifically the pancreas either does not produce insulin at all or it does not effectively use the produced insulin.
- Pancreas gland regulates the level of blood sugar by producing two hormones:
 - ✓ Glucagon that raises the level of blood sugar in the body and
 - ✓ Insulin that lowers the level of blood sugar in the body.
- ❑ Symptoms of Diabetes: The signs and symptoms of diabetes are:
 - (1) Being very thirsty
 - (2) Urinating often
 - (3) Feeling very hungry
 - (4) Feeling very tired
 - (5) Feelings of pins in your feet
 - (6) Losing weight without trying
 - (7) Sores that heal slowly
 - (8) Dry, itchy skin
 - (9) Losing feeling in your feet and Blurry eyesight

Types of Diabetes:

- The three main types of diabetes are type 1, type 2, and gestational diabetes.
- 1. **Type 1 Diabetes (IDDM)**, which used to be called juvenile diabetes, develops most often in young people; however, type 1 diabetes can also develop in adults.
- In type 1 diabetes, your **body no longer makes insulin or enough insulin because the body's immune system**, which normally protects you from infection by getting rid of bacteria, viruses, and other harmful substances, has attacked and destroyed the cells that make insulin.
- Treatment for type 1 diabetes :
 - (1) Taking injections of insulin
 - (2) Sometimes taking medicines by mouth
 - (3) Making healthy food choices
 - (4) Being physically active
 - (5) Controlling your blood pressure levels and Controlling your cholesterol levels

- **2. Type 2 Diabetes (NIDDM)** , which used to be called adult-onset diabetes, can affect people at any age, even children.
- However, type 2 diabetes develops most often in middle-aged and older people.
- People who are overweight and inactive are also more likely to develop type 2 diabetes.
- Type 2 diabetes usually begin with insulin resistance—a condition that occurs when fat, muscle, and liver cells do not use insulin to carry glucose into the body's cells to use for energy.
- As a result, the body needs more insulin to help glucose enter cells.
- Type 2 diabetes is the most common one, which accounts for about 90 percent of all diabetic cases.

- **Treatments for type 2 diabetes:**
- **(1) Using diabetes medicines (2) Making healthy food choices (3) Being physically active (4) Controlling your blood pressure levels and Controlling your cholesterol levels**

3. Gestational Diabetes:

- **Gestational diabetes can develop when a woman is pregnant. Pregnant women make hormones that can lead to insulin resistance.**
- **Overweight or obese women have a higher chance of gestational diabetes.**
- **Also, gaining too much weight during pregnancy may increase your likelihood of developing gestational diabetes.**
- **However, a woman who has had gestational diabetes is more likely to develop type 2 diabetes later in life.**
- **Babies born to mothers who had gestational diabetes are also more likely to develop obesity and type 2 diabetes.**
- **If left untreated diabetes causes complications that lead to blindness, kidney failure, heart diseases, nerve damage and amputations.**

- **3. Asthma:** Asthma is a chronic lung disease characterized by episodic bouts of wheezing, coughing, and difficulty breathing. An asthmatic attack is usually triggered by allergens (e. g., pollen, certain foods, pets, etc), irritants (e. g., cigarette smoking, and smog), exercise or emotional stress and changes in temperature which results in the narrowing of the airways in the lungs. This reaction increases the resistance to the airflow in and out of the lungs, making it harder for the individual to breathe. The severity of asthma varies greatly from mild to profound in which the child may experience only a period of mild coughing or extreme difficulty in breathing that requires emergency treatment. Many asthmatic children experience normal lung functioning between episodes.

- Asthma is the most common lung disease of children, estimates of its prevalence range from 6% to as high as 12% of school-age children.
- **The causes of asthma are not completely known**, though most consider it the result of an interaction of heredity and environment.
- Asthma tends to run in families, which suggests that **an allergic intolerance to some stimulus may be inherited**.
- Primary treatment for asthma begins with **a systematic effort to identify the stimuli and environmental situations that provoke attacks**.
- The number of potential allergens and irritants is virtually limitless, and in some cases it can be extremely difficult to determine the combination of factors that result in an asthmatic episode.
- Changes in temperature, humidity, and season are also related to the frequency of asthmatic symptoms.
- Asthma is the leading cause of absenteeism in school.
- **Chronic absenteeism makes it difficult for the child with asthma to maintain performance at grade level**, and homebound instructional services may be necessary. The majorities of children with asthma who receive medical and psychological support, however, successfully complete school and lead normal lives.

2.3.5 Assessment and identification of children with physical and health impairments:

- The assessment and identification process regarding people with physical and health impairments should include the following specific areas. The process must be done by employing a **multi-disciplinary approach; it should be an ongoing process; and it considers a holistic nature of the child.**
- **A. Activities of daily living:** it involves assessment of current and potential skills in self-help and daily living such as eating, toileting, personal hygiene, cooking, travel and public transportation.
- **B. Mobility:** it focuses on assessing the student's current ability and potential to move from place to place and to become independent.
- **C. Psycho-social development:** it deals with assessing factors that interfere with social and emotional development and the student's ability.
- **D. Communication:** it deals with evaluation of student's ability to understand and express language and his/her ability to communicate.
- **E. Academic potential:** Modifications in the physical setup, elimination of time tasks, or alternative response modes such as verbal rather than written responses are some of the required adaptations.
- **F. Transitional skills:**

- Some of them are:
- **(i) Prosthesis is an artificial replacement for a missing body part such as an artificial leg (limbs).**
- **(ii) Orthosis is a device that enhances partial functioning of a body part such as leg brace.**
- **(iii) Adaptive device is an ordinary item found in the home, office, or school that can be modified.**

2.3.6 Intervention of children with physical and health impairments

- 1. **Hand-on-therapy:** Physical occupational and other specialists provide direct hands on treatment.
- 2. **Assistive devices:** Braces and splints (usually made of molded plastic) are used to give a child movement with stability, to correct abnormal postures, and to control involuntary motions.
- 3. **Medication:** Medication can sometimes help to reduce spasticity and rigidity but generally are of only limited usefulness in improving the muscle tone of children with physical disabilities.
- 4. **Surgery:** Orthopedic or neurologic surgery also helps, although this often regards as a last resort in the treatment of physical impairment.
- For the goodness of intervention, however, concentrating on the following points is important.
- **Using special Devices:** Many children with physical impairment/disabilities use special orthopedic devices

- **Using special Devices:**

- **1. Prosthesis:** These are artificial replacement of missing body part (most frequently arm, leg or eye).
- **2. Wheel chair:** This is prescribed by a physician for individuals who are unable to ambulate or for those whose ambulation is unsteady, unsafe or too strenuous.
- **3. Adaptive devices:** These include using special eating utensils; such as forks and spoons with custom designed handle or straps.
- **4. Orthosis:** This is a device that enhances the partial functioning of a body part; such as a long brace.
- **5. Residual functioning:** These are procedures to helping children to use some of their skills and abilities to become more independent (e.g. prosthetics, orthotics, adaptive devices).
- **6. Facilitating communication:** This refers to communication boards and electronic devices with synthesized speech output.
- **7. Writing aids:** These include computers with word processing programs and with keyboard adoption.
- **8. Positioning:** This refers to sitting, standing, or other positioning device which provide adequate support without locking the child into a static position.

Modifying the Environment

1. Changing desk and table tops to appropriate height.
2. Providing a wooden pointer to enable a student to reach the upper button on an elevator control panel.
3. Adding adaptive devices (rubber bands, plastic wedges, plastic tubing) to writing instruments to make them easier to grasp/grip.
4. Using study buddies in activities that require extensive writing.
5. Installing paper cup dispensers near water functions so that they can be used by students in wheelchair.
6. Using word processors, computers, typewriters, and calculators rather than handwriting responses.
7. Moving a class or activity to an accessible part of a school building; so that a student with a physical impairment can be included.

2.4. Communication Disorders (CDs)

- Communication entails **receiving, understanding and expressing information, feelings and ideas.**
- It is an integral part of our daily lives that most of us take our ability to communicate for granted.
- Most children come to school able to understand others and express themselves.

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- Their communication abilities allow them to continue developing socially and take part in the lives and activities of their academics.
- However, some others do have communication problems that adversely affect their social interactions and academic performance.
- CDs are difficulties in **communication and language usage** that include a number of speech problems (such as **articulation disorders, voice disorders, and fluency disorders**) and **language problems** (such as **difficulties in receiving information and expressing language**).

2.4.1 Classifications of Communication Disorders

- broadly classified into two broad categories:
Speech disorders and language disorders.
- 1) **Speech Disorder** difficulty producing sounds as well as disorders of voice quality or fluency of speech.
- **Speech disorders include disorders of articulation, voice and fluency (rate and rhythm of speech).**

Cont...

- **Articulation Disorders** are one of the most frequent communication disorders in preschool and school-aged children.
- Articulation is a speaker's production of individual or sequenced sounds.
- Articulation disorders, therefore, are problems in producing sounds correctly.

Cont...

- Articulation errors or problems can be manifested in the form of substitutions, omissions, additions and distortions.
1. **Substitutions:** - these are common, as when children substitute d for the voiced th (“doze” for “those”) t for c (“tat” for “cat”), or w for r (“ Wabbit” for “rabbit”).
 2. **Omissions:** - these occur when a child leaves a phoneme out of a word. Children often omit sounds from consonant pairs (“boo” for “blue” “cool” for “school”) and from the ends of words (“ap” for “apple”).
 3. **Distortions:** - these are modifications of the production of a phoneme in a word; a listener gets the sense that the sound is being produced, but it sounds distorted. Common distortions, called lisps, occur when s,t,sh, and ch are mispronounced.
 4. **Additions:** - occur when children add extra sounds, making comprehension different or they occur when children place a vowel between two consonants, converting “tree” into “tahree”, “buhrown” for “brown”.

Cont...

- 2) Voice Disorders are reflected in speech that is hoarse, harsh, too loud, and too high pitched, or too low pitched.**
- ✓ Each person has a unique voice. This voice reflects the interactive relationship of pitch, duration, intensity, resonance and vocal quality.**

Cont....

- **Pitch disorders:** - it involves too high or too low pitch, or a pattern of pitch that is so monotonous that it calls attention to itself.
- **Disorders of loudness:** - in this case speech is habitually too loud or soft for the setting, or that sporadic bursts or cycles of excessive loudness are observed.
- **Voice quality disorders:** - this involves breathiness, harshness and nasality

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- **Fluency Disorders:** Normal speech requires correct articulation, vocal quality, and fluency (rate and rhythm of speaking).
- Fluent speech is smooth, flows well, and appears effortless.
- Fluency disorders are characterized by interruptions in the flow of speaking, such as a typical rate or rhythm as well as repetitions of sounds, syllables, words, and phrases.
- They often involve what is commonly called “Stuttering”, Stuttering occurs when a child’s speech has a spasmodic hesitation, prolongation, or repetition.

2) Language Disorder

- It is a difficulty in **receiving, understanding, and formulating ideas and information.**
- Therefore, language disorders are manifested in a significant impairment in **a child's receptive or expressive language or both.**

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- **Receptive language (RL)** involves the **reception and understanding of language**. CW a RLD has a malfunction in the way they receive information. Information comes in, but the child's **brain has difficulty processing it effectively**.
- **Expressive language (EL)** involves the **ability to use language to express one's thoughts and communicate with others**. A problem in speaking is a common expressive language disorder.

- Generally language impairments adversely affect;
 - ✓ A Student's Social Skills And
 - ✓ Academic Performance.
- It includes;
 - ❖ **Phonology**
 - ❖ **Morphology**
 - ❖ **Syntax**
 - ❖ **Semantics**
 - ❖ **Pragmatics**

- **Phonology:** - it refers to the use of sounds to make meaningful syllables and words. Students with phonological disorders may be unable to discriminate differences in speech sounds or sound segments that signify differences in words. For example, to them the word “pen” may sound no different from “pin”. Their **inability to differentiate sounds as well as similar rhyming syllables may cause them to experience reading and/ or spelling difficulties.**
- **Morphology:** - it is the system that governs the structure of words. Children with morphological difficulties have problems using the structure of words to get or give information. They may make a variety of errors. For example, they may not use “-ed” to signal past tense as in “walked” or “-s” to signal plurality.
- **Syntax:** - it provides rules for putting together a series of words to form sentences. Syntactical errors are those involving word order, such as ordering words in a manner that does not convey meaning to the listeners.
- **Semantics:** - it refers to the meaning of what is expressed. Semantic development has both receptive and expressive components. Children who experience difficulty-using words singly or together in sentences may have semantic disorders. They may have difficulty with words with **double meanings**, abstract terms, synonyms and idioms.
- **Pragmatics:** - it refers to the use of communication in contexts. It focuses on the social use of language—the communication between a speaker and listener within a shared social environment.

2.4.2 Major Causes of Communication Disorders (CDs)

- The causes of **CDs** can be classified into two:
Classification by the Cause.
- According to this criterion, causes of CDs are classified into two:
(1) **organic disorders**; and
(2) **functional disorders.**

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- **Organic disorders** caused by an **identifiable problem in the neuromuscular mechanism of the person**. The causes of organic disorders are numerous; they may originate in the **nervous system, the muscular systems, the chromosomes, or the formation of speech mechanism**.
- They may **include hereditary malformations, prenatal injuries, toxic disturbances, tumors, traumas, seizures, infectious diseases, muscular diseases and vascular** impairments.
- **Functional disorders** CDs with **no identifiable organic or neurological cause**. A functional speech and/or language disorder is present when the **cause of the impairment is unknown**.

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- **Classification by the Onset:** According to when the problem began, causes of CDs can be classified into two:
- (1) **Congenital Impairment** (an impairment that occurs at or before birth); and
- (2) **Acquired Impairment** (an impairment that occurs after birth).

2.4.3 Support Systems of Children with Communication Disorders:

- Since language and speech are **integral parts of academic and social development.**
- **Team approach, involving the classroom teacher, the speech language pathologist, parents, and other professionals** is essential in intervention and management of CDs

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- **Treating Speech Sound Errors:** A general goal of specialists in CDs is to help the child speak as clearly and pleasantly as possible so that a listener's attention will focus on the child's message rather than how he/she says.

- **Articulation Errors:** - the goal of therapy for articulation problems are acquisition of the correct speech sound(s), generalization of the sound(s) to all speaking settings and contexts, and maintaining of the correct sound(s) after therapy has ended. This therapy includes: discrimination activities and production activities.
- Discrimination activities are designed to improve the child's ability to listen carefully and detect the differences between similar sounds (e. g., the /t/ in "take", the /c/ in "cake") and to differentiate between correct and distorted speech sounds. The child learns to match his/her speech to that of a standard model by using auditory, visual and tactual feedback.
- A generally consistent relationship exists b/n children's ability to recognize sounds and their ability to articulate them correctly.

- **Phonological Errors:** - when the child's spoken language problem includes one or more of phonological errors, **the goal of therapy is to help the child identify the error pattern(s) and gradually produce more linguistically appropriate sound patten.** For example, a child who frequently omits final consonants might be taught to recognize the difference between minimally contrastive words, such as "sea", "seed", "seal", "seam", and "seat"

- **Treating Fluency Disorders:** For many years, stuttering was widely thought that a tongue that was unable to function properly in the mouth caused it. As a result, it was common for **early physicians to prescribe ointments to blister or numb the tongue or even to remove portions of the tongue through surgery.** However, in recent years, application of behavioral principles for treating fluency disorders has got strong influence. A therapist using this methodology regards **stuttering as learned behavior and seeks to eliminate it by establishing and encouraging fluent speech. This can be done by positively reinforcing the child's fluent utterances both at home and in school.**

- **Treating Voice Disorders:** Voice therapy is often used to teach the child with the problem to listen to his/her own voice and learn to identify those aspects that need to be changed.
- Depending on the type of voice disorder and the child's overall circumstances, vocal rehabilitation may include activities such as **exercises to increase breathing capacity, relaxation techniques to reduce tension, or procedures to increase or decrease the loudness of speech.**
- Because many **voice problems are directly attributable to vocal abuse, behavioral principles can be used to help children and adults break habitual patterns of vocal misuse.**
- For example, a child might self-monitor the number of abuses he/she commits in the classroom or at home, receiving reinforcement for gradually lowering the number of abuses over time.

- **Treating Language Disorders:** This therapy focuses on pre-communication activities that encourage the child to explore and that make the environment conducive to the development of both receptive and expressive language. In addition, because children learn through imitation, it is important for the teacher or specialist to talk clearly, use correct inflections, and provide a rich variety of words and sentences.
- Speech-language pathologists are increasingly employing naturalistic interventions to help children develop and use language skills.
- Naturalistic interventions should be provided: When the child is interested for the lesson, teach what is functional for the student at the moment.

Unit Three: Cognitive Impairments/Disabilities

- 3.1 Intellectual Impairment/Disability (II or ID)
- 3.1.1 Definition of Intellectual Impairment (Disability)
- The 1992 AAMR'S DEFINITION:
- “Mental retardation refers to a disability characterized by **significantly sub-average general intellectual functioning**, existing concurrently with related **limitations in two or more of the following applicable adaptive skill areas**(communication, self-care, home living, social skills, community use, self-direction, health and safety, functional academics, leisure, and work). Mental retardation **manifests before age 18.**”

- **2. The 2002 AAMR's Definition:** In 2002, the AAMR published a definition and approach for diagnosing and classifying mental retardation that represented a conceptual shift from **viewing intellectual disability as an inherent trait or permanent condition to a description of the individual's present functioning and the environmental supports needed to improve it.**
- **"Mental retardation is a disability characterized by significant limitations both in intellectual functioning and in adaptive behavior as expressed in conceptual, social, and practical adaptive skills. This disability originates before age 18".**

3.1.2 Classification of Intellectual Impairment/Disability (II)

No	Degree of Severity of ID	Educational Characteristics CWID	Intensity of support needed
1	Mild-ID (50/55-70 IQ scores	Educable: Children with this category; they are: ☐ not noticeably d/t from the “normal” ones in their physical characteristics and general health and capable of learning fundamental academics and personal responsibilities. ☐ also able to function within the traditional level curriculum with only minor modification and assistance and be self-sufficient and live independently as productive members of the community as adults. ☐ delayed 1 to 3 years in school performance	Intermittent support is given when necessary eg. First day of a school, job, and loss of as well as they face health crisis

No	Degree of Severity of ID	Educational Characteristics CWID	Intensity of support needed
2	Moderate-ID (35-49 IQ scores)	Trainable: Children in this category have/are: ❖ seriously impaired adaptive capacities and a functional ability of nearly 1/2-1/3 of expected for their chronological age. ❖ able to master self-care skills, basic language, and cognitive concepts. ❖ be able to live in community homes and work with supervised workshop facilities as adults and opportunities to benefit from vocational, occupational and social trainings ❖	Limited support is given for a long period with specific time eg. 3 months of training before starting a job

No	Degree of Severity of ID	Educational Characteristics CWID	Intensity of support needed
3	Severe-ID (20-34 IQ scores)	Supportable: Children in this category ☐ manifest poor motor development and little or no speech before age 7 ☐ may learn to talk during their later school years and can learn basic hygiene skills and profit little from vocational training. ☐ may be able to perform simple tasks with close supervision	Extensive support is given regularly at least in one place

No	Degree of Severity of ID	Educational Characteristics CWID	Intensity of support needed
4	Profound-ID (< 20 IQ scores)	Life-support: Children in this category ☐ manifest only minimal sensory motor functioning before their age 7. ☐ may show some motor development during their latter school years, and may benefit minimal from self-care training. ☐ may develop some speech, able to take a very limited self-care as adults ; ☐ require constant supervisions in a very structured environment.	Pervasive support is given regularly in all or across all environments

3.1.3 Major Causes of Intellectual Impairments/ Disabilities

- **3.1.3.1 Genetic Causes**
- **Genetic conditions** abnormalities of genes inherited from parents, errors when genes combine, or from other disorders of the genes caused during pregnancy by infections, overexposure to x-rays and other factors. There are
 - **A. Chromosomal Abnormalities/Genetic Defects:** There are a number of forms of intellectual disability that are caused by genetic factors. *Examples:*
 - 1. **Down's syndrome (DS)** is an example of a chromosomal disorder which is caused by too many or too few chromosomes, or by a change in structure of a chromosome. The vast majority of cases are caused by an **extra copy of chromosome 21, with translocation** (when part of this chromosome attaches to another chromosome) and **mosaicism** (when only some cells include an extra copy of chromosome 21) representing less frequent causes.
 - **Down's syndrome** is one of the most common chromosomal disorders, accounts for the largest number of cases of intellectual disability in which extensive and pervasive supports are required. It is caused by the presence of extra chromosomal materials in the cells.
 - The most common form is called **Trisomy 21** because of an **extra chromosome attached to the twenty-first chromosome** pair. The risk of giving birth with Down's syndrome increases with maternal age after 30 or 35 years.
 - 2. **Fragile X-Syndrome (FXS)** is a single gene disorder located on the X chromosome and is the leading inherited cause of intellectual disability. FXS is a common hereditary cause of ID associated with **genetic anomalies in the 23rd pair of chromosomes**. Although less prevalent than DS, FXS is the most common known inherited cause of ID, full mutation of the Fragile X Mental Retardation-1 gene (*FMR1*). This causes a **deficiency in production of the Fragile X Mental Retardation Protein (FMRP)**, which is thought to be essential for normal cognitive functioning. Common in males

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- **B. Metabolic Disorders:** characterized by the body's **inability to process (metabolize) certain substances that can then become poisonous and damage tissue in the central nervous system.** The most frequent and best known forms of intellectual disability resulted from metabolic is **Phenylketonuria (PKU).** **PKU** is a condition caused by the inheritance of two recessive genes from parents who are carriers of the condition.

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- PKU gene resulted in the **lack of the production of an enzyme that processes proteins, there is a building of an acid called phenylpyruvic acid.** This building causes brain damage.
- In other words, with PKU, the baby is not able to process phenylalanine, **a substance found in many foods, including the milk ingested by infants.**
- The inability to process **phenylalanine** results in an accumulation of poisonous substance in the body. If it goes untreated or is not treated properly (usually through dietary restrictions). This damage leads to **severe or profound intellectual disability.**
- **Broad screening programs and diet therapy may reduce the disorder.**

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- The second common example of **metabolic disorder** is **Galactosemia**. Galactosemia is a metabolic disorder that causes an **infant to have difficulty-processing lactose**.
- **galactosemia**, the child is unable to properly process lactose, which is the primary sugar in milk and is also found in other foods.
- If galactosemia remains untreated, **serious damage results, such as cataracts, heightened susceptibility to infection, and reduced intellectual functioning**.
- **Dietary controls** must be undertaken, eliminating milk and other foods containing lactose.

C. Hormonal Disorders; for example, **Hypothyroidism/Cretinism** is the best example that can be included under hormonal disorder that may lead to the development of ID.

Here intellectual disability results from **dysfunction of thyroid gland**.

Most children with this disorder suffer from severe retardation.

However, good results can be obtained if early intervention is made.

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- **3.1.3.2 Prenatal, Peri natal and Postnatal Causes:** Genetic and chromosomal causes of intellectual disability are present from the moment of conception. Pre, peri and post-natal causes.
- **Problems during pregnancy (prenatal risk factors):** - Use of alcohol or drugs by the pregnant mother can cause intellectual disability. In fact, **alcohol, smoking** in increasing the risk of intellectual disability. Other risks include **malnutrition, certain environmental toxins, and illnesses of the mother during pregnancy, Rh factor, blood incompatibility, infectious disease such as toxoplasmosis, cytomegalovirus, rubella and syphilis**, etc.
- **Problems at birth (perinatal risk factors):** - **Prematurity and low birth weight** predict serious problems
- Difficulties in the birth process such as **temporary oxygen deprivation, complicated delivery, prolonged delivery, infectious diseases, infections or birth injuries** may cause intellectual disabilities.
- **Problems after birth (postnatal risk factors):** - **Lead, mercury and other environmental toxins** can cause irreversible damage to the brain and nervous system. **Infectious diseases, child abuse and neglect, accidents, and the like may result in brain damage** of the fetus and consequent retardations.

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- **3.1.3.3 Cultural-Familial Causes:** Cultural-Familial Cause is another major variable of intellectual disability. **It is the causes of 75% of the retarded children for whom there is no organic pathology and whose retardation is presumed to be due to the combination of hereditary and environmental factors.**
- Absence of good reinforcement for intelligent behavior, **harsh punishment for specific behaviors, homelessness, poverty, severe abuse and neglect, malnutrition, psychosocial deprivations** are some **environmental factors that can cause intellectual disability.**
- Generally, intellectual disabilities that may be attributable to both socio-cultural and genetic factors is ***cultural-familial intellectual disabilities***.
- People with this condition are often described as **(1) having mild intellectual disabilities, (2) having no known biological cause for the conditions, (3) having at least one parent or sibling who have mild intellectual disabilities, and (4) growing up in a low socio-economic-status or deprived and impoverished environments.**

3.1.4 Characteristics of Children with ID

A) Cognitive/ Intellectual Characteristics: **CWID** have limitations on the following intellectual functioning.

1. Memory problem:-PWID have impairments in memory, especially in short-term memory. The problem is said to be due to the inability to use good learning strategies such as grouping items and rehearsal.

2. Transference problem in learning: PWID are said to experience difficulties in applying information to new situations

3. Low Motivation: This low motivation leads them to a problem solving style that is called **outer-directedness** - distrusting one's own solutions and looking excessively to others for guidance.

4. Imitation problem: is a skill, which children develop naturally through play and learning experiences at home and with peers. Many PWID have imitation problem and need specific instruction and practice before they can imitate a model.

5. Difficulty in discrimination: CWID are usually in trouble to discriminate between shape, color, size, brightness and positions.

6. Attention problem: CWID have attention deficits. It is manifested in three areas: short attention span, problem of focusing attention, and problem of selective attention.

7. Adaptive behaviors: CWID have limitations in skills needed to function in different environments.

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- **B) Educational/Academic/ Characteristics:** SWLDs show a discrepancy b/n their score on intelligence and achievement tests
- reading comprehension, in mathematical reasoning, and in basic statistics such as **addition, subtraction, multiplication, and division.**
- Professionals in the area pointed out that **repeated failure and frustrations in classroom, and inappropriate expectations and tasks are the leading causes of low performance.**
- **C) Social/Emotional Characteristics:** social and behavioral problems.
- They show **socially inappropriate behaviors** from the normal ones. They lack social skills to establish and maintain friendship. Because they are both socially and emotionally immature, behavioral problems such as **disruptiveness, attention deficits, poor self-image or self-concept, rigidity, distractibility**

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- **D) Physical/Motor Characteristics: CWID**
- lag behind their normal peers on measures of **gross motor proficiency, and physical fitness.**
- They show **a marked difference in body coordination, strength, and flexibility.**
- According to Garwood (1983), *(1) Sequential patterns of movement (2) Keeping body balance (3) Basic physical abilities (i.e., strength, flexibility, agility, and endurance) (4) Fine motor activities that are timed (5) Eye-hand coordination (6) Eye- foot coordination and (7) Dexterity*
- **E) Language and Communication Characteristics:** are very much related and positively correlated. Individuals who show delay in cognitive functioning typically **show delay in language and communication skills.**
- The most common speech problems associated with CWID are **defects in articulation, voice and stuttering.**

3.1.5 Intervention of CWID

- Special assistance and care are essential for PWID to enable them lead better and sustainable life. Accordingly, the intervention strategies should consider the following two areas:
- **A) Family Support:** parents of CWID face many challenges to **overcome learning problems and promote optimal development** for their children. Hence, parents need **training, an intensive counseling, and proper guidance** as early as possible. Furthermore, **parents need to closely work with professionals.**
- **B) School Support:** School supports based on their level of ID. Teachers also need to be **trained with the necessary skills** that help them train, treat and manage CWID.
- Teachers should use the following **SEVEN TIPS** while they are teaching/training children with intellectual disability. They are:
- **1) Using multimodalities (2) Using concrete, varied and familiar examples (3) Providing supportive and corrective feedback (4) Using appropriate assessment (5) Making instruction interesting (6) Breaking lessons in to smaller parts and (7) Employing repetitions**

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- **C) Behavioral techniques** (such as **operant conditioning**), and **psychotherapy** especially for severely and profoundly intellectual disability.
- **Prevention:** - Prevention can be primary, secondary, or tertiary.
 - ✓ Primary prevention is taken **to prevent the problem from its occurrence.**
 - ✓ Both secondary and tertiary preventions mainly **focus on minimizing the severity of the problem.**
 - The following are samples of preventive measures at different periods.
 - ❖ **Before pregnancy:** *good immunization programs, and genetic counseling regarding inheritable disorders*
 - ❖ **During pregnancy:** *good nutrition, avoidance of drugs, toxins and the like*
 - ❖ **After birth:** *immunizing the child, good and accessible medical care, balanced diet, and the like can reduce the incidence and severity of retardation among children.*

3.2 Learning Disabilities (LDs)

- **3.2.1 Definition of Learning Disability:** The U.S. National Joint Committee on LDs (NJCLD, 1988) defined LDs follows: “LDs is a general term that refers to **a heterogeneous group of disorders manifested by significant difficulties in the acquisition and use of listening, speaking, reading, writing, reasoning, or mathematical abilities.**
- These disorders are intrinsic to the individual, presumed to be due to **central nervous system dysfunction and may occur across the life span.**
- Although **LDs may occur concomitantly with other disabilities (e.g., sensory impairment, mental retardation, serious emotional disturbance), or with extrinsic influences (such as cultural differences, insufficient or inappropriate instruction),**
- **They are not the result of those conditions or influences”**

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- From the definition given above, the following main points can be inferred:

- ❑ ***The problem is heterogeneous.*** This means that if we have 20 students with LDs, we find that the problem manifest itself in 20 different ways and LDs occur across life span. This implies that the problem occurs throughout an individual's life.
- ❑ ***The cause of LDs is intrinsic to the individual most likely due to CNS dysfunction. .***
- ❑ ***LDs are not a problem WI.*** This means that people with LDs have average and above average ***IQ and they seem to be bright. However, they are unable to perform and achieve as expected (up to their potential).***
- ❑ Generally, though most ***IWLDs*** are as smart as their normal peers, they see, hear, read, process and understand things differently!!

3.2.1.2 CRITERIA FOR DEFINING AND LABELING OF LDS

- a. Inclusionary criteria:** - LDs must include disabilities related to speaking, writing, reading, calculating, listening, spelling and thinking.
- b. Exclusionary criteria:** - LDs are not primarily due to impairments arising from other disabilities, such as sensory disabilities, EBDs, ID, etc.
- c. IQ-Achievement discrepancy:** - Though people with LDs have normal and above normal intelligence, they are not able to achieve as expected.
- d. Presumption of CNS dysfunction:** - Though the cause of LDs is presumed to be due to CNS malfunctioning, it does not mean that there is tissue damage.
- e. Psychological processing disorders:** - Most IWLDs have difficulties in gathering, perceiving, processing and interpreting visual and auditory stimuli.
- f. Need criteria:** The student needs special education. That means without specialized supports, the disability prevents him/her from learning and benefiting.

3.2.2 Causes of Learning Disabilities

A) **Genetic or Heredity Factors:** LDs are genetically determined with genetic traits manifesting themselves via the inheritance process. The concordance rate is higher for identical than fraternal twins for reading disorder. Some findings suggest that genes on **chromosome 15 or 6 may cause reading disorder** for a minority of individuals.

B) **Neurological Factors:** LDs is a reflection of structural damage or improper development of the NS. Such problems could occur **during prenatal or postnatal period, as the NS is developing.**

❑ Alternatively, **head injury, lack of oxygen, exposure to toxins, seizures and nutritional deficiencies may contribute to central nervous system damage.** Various neuropsychological deficits are associated WLDs, such as in **visual spatial perception, auditory perception, semantic memory and phonemic discrimination.** CW reading Ds have left-hemisphere deficits and autonomic Ds such as **eczema, allergies.**

C) **Environmental Factors:** can indirectly play a role in causing LDs.

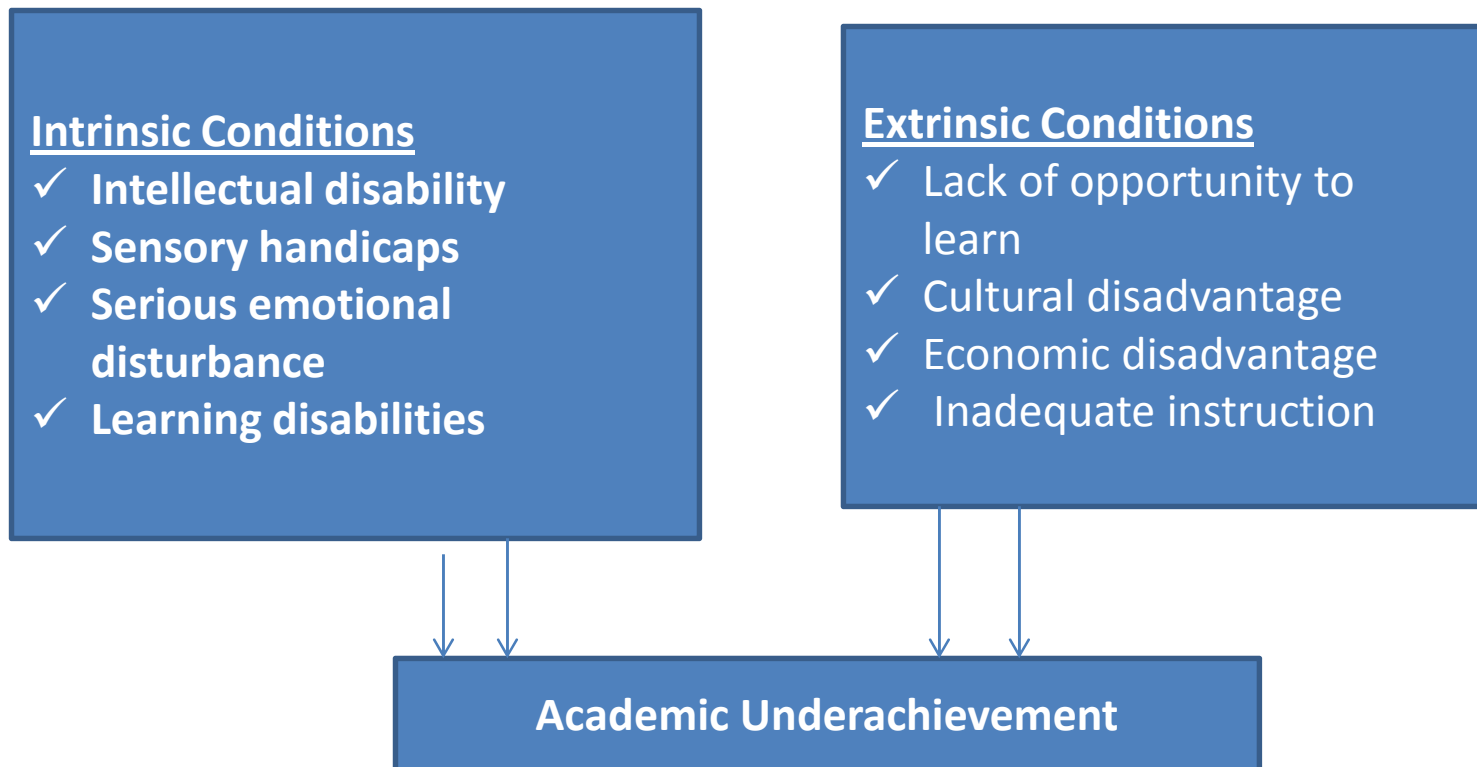
❑ factors such as **malnutrition, prematurity, poor prenatal and post-natal health care, stress, poor parenting and teaching** can have a negative impact on learning by creating conditions in which brain dysfunction is more likely.

❑ **Substance abuse such as alcohol and other drugs, defective learning models, emotional disturbances, social and cultural deprivation can contribute to learning disabilities.**

❑ **poverty and inappropriate and/or inadequate stimulation on the development of learning disorders in children.**

3.2.3 LDs, Other Disabilities, and Academic Achievements

- Learning disabilities are one of the many factors, which could cause academic underachievement.
- The relationship among learning disabilities, other disability areas, and academic achievement can be presented in the diagram



3.2.4 Prevalence of Children With Learning Disabilities

- SWLDs is increasing at an alarming rate when compared to other areas of exceptionality.
- More specifically, SWLDs account for 50% of all students with other disabilities.
- Reason dramatic increase in the number of SWLDs. *(1) Better research done by specialists concerning the problem (2) Increased awareness of parents and teachers about the problem (3) LDs gained full acceptance from government, parents, professionals, schools, etc.(4) Children who were once misdiagnosed as having ID are now being recognized as having LDs (5) Programs for LDs have been expanded to include preschool children (6) Improvements in procedures of identification and assessment process of individuals with LDs and (7) Males are outnumbered females in their prevalence*

3.2.5 Types of Specific Learning Disabilities

A) Learning disabilities in reading (dyslexia):. Signs of reading difficulty include problems with: **(1) Letter and word recognition (2) Understanding words and ideas(3) Reading speed and fluency (4) General vocabulary skills**

B) Learning disabilities in mathematics (dyscalculia): **a visual disorder or a difficulty with sequencing, memory or organization.**

A child with a math-based learning disorder may **struggle with memorization and organization of numbers, operation signs, and number “facts”** (like $5+5=10$ or $5 \times 5=25$).

Children with math learning disorders might also have **trouble with counting principles (such as counting by twos or counting by fives) or have difficulty telling time.**

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- **C) Learning disabilities in writing (dysgraphia):** LDs in They include problems with: **(1) Neatness and consistency of writing (2) Accurately copying letters and words (3) Spelling consistency and (4) Writing organization and coherence**
- **D) Learning disabilities in motor skills (dyspraxia):** Motor difficulty (MD) **problems with movement and coordination** whether it is with fine motor skills (cutting, writing) or gross motor skills (running, jumping).
- It is “output” activity meaning that it relates to the output of information from the brain.
- In order to run, jump, write or cut something, the brain must be able to communicate with the necessary limbs to complete the action. Signs that your child might have a motor coordination disability include problems with physical abilities that require **hand-eye coordination, like holding a pencil or buttoning a shirt.**

3.2.6 Interventions: Helping Students with LDs Learn Best

- **Improving memory problems:** problems in their short-term memory. For these children the following treatment techniques;
 - (1) Reducing verbal distractions**
 - (2) Using varied and familiar examples**
 - (3) Making tasks interesting**
 - (4) Decreasing time duration for a task**
 - (5) Using multimodalities**
 - (6) Maintaining an eye contact with students**
 - (7) Giving short assignments and tests**
 - (8) Encourage a child to use study skills**
 - (9) Grouping items into meaningful units**
 - (10) Encourage the child to use mnemonics**
 - (11) Teach the child to apply rehearsal and**
 - (12) Apply well-organized, clear and sequential instruction**

Cont...

- B) Improving writing problems: includes **handwriting practice, direct practice in spellings, sentence writing and review of grammatical rules.** Spelling problems frequently involve **deficits in phonemic awareness and word recognition;**
- Spelling skills can be promoted by teaching the child spelling rules *e.g.*, before *e* except after *c*, **spelling of word roots, prefixes and suffixes** and **mnemonic strategies for remembering the spelling of specific irregular words.** **Spelling games and repetition learning are used for children with spelling deficits.**

Cont...

- Finally the child gradually says the words more and more silently until the child is writing while thinking (but not saying the words).
- ✓ **Guided practice,**
- ✓ **Allowing note taker,**
- ✓ **Using audio recorder,**
- ✓ **Videotape reports and ready made notes can also help a child** to improve his writing difficulties.

Cont...

- **C) Improving mathematics problems:** teaching mathematics concepts with continuous practice in solving math problems.
- Once these errors are identified several techniques may be used to help the child.
- ✓ **Concrete objects,**
- ✓ **pictorial representations** such as graphs, analogies and logical explanations are used to make abstract concepts meaningful.
- ✓ **Learning math rules** (*e.g.*, BODMAS) and computer games may be taught.
- ✓ **Using visual pictures** (drawing pictures of word problems),
- ✓ **use of memory aids** (music) and use a computer for drills and practice are among the techniques that can be used to help a child remember math concepts.

D) Improving reading difficulties:

- ❖ vocabulary-building exercises,
- ❖ sight reading exercises,
- ❖ learning word roots, prefixes and suffixes.
- ❖ Teaching sounds by combining consonants and vowels till they become automatic,
- ❖ encouraging them to summarize materials and use self – questioning while reading,
- ❖ using continuous timed reading practice,
- ❖ giving students extra time to finish tasks,
- ❖ providing taped tests that allow the child to hear the questions instead of reading them

E) Improving speaking problems: exhibit difficulties both in the **mechanical and social use of language.**

The following interventions are recommended.

- **Modeling**(saying the correct one) and reinforcing the correct attempt of the student
- **Giving different contexts so that students with learning disabilities can practice the language**
- **Allowing students to summarize texts/passages read by the teacher**

3.3 Emotional and Behavioral Disorders

- **3.3.1 Definition of Emotional And Behavioral Disorders:**
- It is difficult to have a precise and universally acceptable /common definition regarding EBDs. This is because of:-
 - ✓ **Lack of precise definitions of normal and abnormal B** as a child's behavior exists along a continuum.
 - ✓ **A cultural d/c as a B is culture specific.** In other words, what is normal in one culture may not be normal in another culture.
 - ✓ **Differences among conceptual models** as each perspectives tries to define a deviant B based on its philosophy.
 - ✓ **Relationship b/n EBDs and other Ds** as there are some similarities in terms of characteristics and effect.
 - ✓ **Lack of objective and precise data gathering tools and study design about EBDs.**

Cont...

- For defining and labeling a child as having EBD, the following criteria must be fulfilled.
 - 1. The problem must occur persistently for at least six months.** The child's behavior must be all three, i., e., **severe** (means the behavior must hamper a normal functioning to a significant degree),
 - 2. The problems must occur across contexts.** The behavior must occur in school and in at least one other setting.
 - 3. The problem must occur to a marked extent that it calls the attention of others.** This means that the behavior must be significantly deviated from generally accepted, age appropriate, ethnic or cultural norms.
 - 4. The problem must adversely affect a child's educational performance and social life.**

- Therefore, by taking in to account the above mentioned definition criteria,
 - 1) An **inability to learn that cannot be explained by intellectual, sensory, or other health factors.**
 - 2) An **inability to build or maintain satisfactory interpersonal relationships with peers and teachers.**
 - 3) **Inappropriate types of behavior or feelings under normal circumstances.**
 - 4) **A general pervasive mood of unhappiness or depression.**
 - 5) **A tendency to develop physical symptoms or fears associated with personal or school problems.**

3.3.2 Characteristic of PWEBDs

- **A) Behavioral Characteristics of CWEBDs externalizing and internalizing.** Both patterns of abnormal behavior have adverse effects on children's academic achievement and social relationships.
- **Externalizing Behaviors:** - are those in which the individual is disturbing to others and include behaviors such as aggression, acting out, and disobedience.
 - (1) *Displays recurring pattern of aggression to ward objects/persons***
 - (2) *Argues excessively***
 - (3) *Destroying their own and others' belongings***
 - (4) *Forces the submission of others through physical and/or verbal means***
 - (5) *Exhibits persistent pattern of tantrums***
 - (6) *Get out of their seats***
 - (7) *Disturb peers***
 - (8) *Hitting and fighting with others***
 - (9) *Frequently exhibits lack of self-control and acting-out behaviors***
 - (10) *Are non-complaints with reasonable requests***
 - (11) *Exhibits persistent pattern of lying and stealing and***
 - (12) *Destroy property, etc.***

Cont...

- **Internalizing Behaviors:** - are those in which the individual him/herself is “disturbed” and include behaviors such as personality problems/act in, anxiety disorders and mood disturbances. The following internalizing behaviors are the major manifestations;
 - (1) Exhibits sad affect, depression, and feelings of worthlessness
 - (2) Has auditory or visual hallucinations
 - (3) Cannot keep mind off certain thoughts, ideas/situations
 - (4) Cannot keep self from engaging in repetitive and/or useless actions
 - (5) Suddenly cries, or displays totally unexpected and a typical affect for the situation
 - (6) Complains of severe headaches, stomach aches, dizziness, vomiting as a result of fear or anxiety
 - (7) Talks of killing self-reports, suicidal thoughts and or is preoccupied with death
 - (8) Shows decreased interest in activities that were previously of interest
 - (9) Shows signs of physical, emotional, and /or sexual abuse
 - (10) Exhibits withdrawal, avoidance of social interactions, and/or lack of personal care

Cont...

- **B) Academic Characteristics:**
- “An inability to learn which **cannot be explained by intellectual, sensory, or health factors.**”
- perform one or more years below grade level academically.
- **students exhibit significant deficiencies** in reading and in math achievement.
- language delays, which compound their difficulties in mastering academic skills and contents.
- They do have a reciprocal relationship between their academic and social behaviors:
- Approximately 50 percent of SWEBDS drop out of school.
- Most cwebds are underachievers in their academic affairs and most of them also exhibit a combination of school failure,
- Antisocial and social withdrawal behaviors.

Cont...

- **D) Social and Interpersonal Relationships:** These children have:
 - low levels of empathy towards others,
 - participate in fewer curricular activities,
 - have less frequent contacts with friends, and
 - they have lower quality relationships than their normal peers.
 - Most SWEBDs have expressive and/or receptive language disorders.
 - they tend to act out their feelings instead. Because of this factor most CWEBDs are underachievers in their academic affairs and
 - most of them also exhibit a combination of school failure, antisocial and social withdrawal behaviors.

3.3.3 Causes of Emotional and Behavioral Disorders;

- **A) Biology:** biological and genetic causes for EBDs are being identified.
- ❑ **Genetic makeup** and children who are born in families where parents or relatives have a history of
- ❑ **Chemical imbalance** is another causal factor of EBDs and this is caused by an **imbalance in dopamine, serotonin, acetylcholine and norepinephrine neurotransmitters** or
- ❑ **Low levels of minerals** such as **Zinc, Magnesium or Potassium in the body**. For example, research now tells us that **mood disorders, depression, and schizophrenia** may have a genetic foundation. Knowing whether biological reasons are part of the cause of a disorder can play a role in treatment. For example, antidepressants are now an important component in many treatment programs for depression.
- ❑ **B) Home and Community:** Environment and culture are the context in which behavior unfolds. No one lives in a social vacuum. Everyone is a member of an immediate family, an extended family, or a community network (neighborhood, church, and clubs).
- ❑ **high rate of negative interactions, lack of interest and concern, and poor adult role models.**

Cont...

- **(i) School:** Teachers' expectations affect the questions they ask students, the feedback they give, and the number and character of their interactions with students.
- Here are some key components of safe and effective schools:
 - (1) Consistency of rules, disciplinary measures**
 - (2) Positive school climate**
 - (3) High level of supervision in all school settings**
 - (4) Good role models from teachers and peers**
 - (5) High levels of parent and community involvement**
 - (6) Well-utilized space and lack of overcrowding**
 - 7) Free from drugs and other environmental hazards**
 - (8) Positive interactions and feedback from peers and teachers**
 - (9) Appropriate instruction delivered and**
 - (10) Rewarding appropriate behaviors and vice versa**

3.3.5 Assessment and Intervention of CWEBDs

A. Teachers and Parents' Observation: should confirm that students are engaging in aggressive, disruptive, inattentive and other socially inappropriate behaviors.

B. Individualized Intelligence Tests: suggests that intelligence of SWEBDs is usually, but not always, in the low average to slow learner range.

C. Individualized Achievement Tests: suggest that the SWEBDs usually, but not always, scores below average across academic areas in comparison to his /her SWOEBDS peers.

D. Behavior Rating Scales: SWEBDs score in the significant range on specific behavioral excesses or deficiencies when compared with others of the same culture and developmental stage.

E. Assessment Measures of Social Skills, Personality, Emotional /or Adjustment: Assessment results here indicate that SWEBDs show significant difficulties in their performance in one or more areas of social skills, personality, emotional and/or adjustment as compared to their non/disabled peers.

3.3.5.2 Intervention:

- (1) **Social skills instruction:** is an important curriculum component for SWEBDs. Many of these children have difficulty holding a conversation, expressing their feelings appropriately, participating in-group activities, and responding to failure or criticism in positive and constructive ways. They often get into fights and altercations because they lack the social skills needed to handle or defuse provocative incidents. Therefore, these students should be taught with the necessary social skills that help them handle their problems in their social life.
- (2) **Academic instruction:** Although SWEBDs require specific education to work on their specific behavior problems and social skills deficits, academic instruction cannot be neglected. Most SWEBDs are already achieving below their nondisabled peers; reading, writing and arithmetic are as important to CWEBDs as they are to any child who hopes to function successfully in society. Hence, if these children are provided with explicit, systematic instruction, they make excellent progress.
- (3) **Behavior Management:** Traditionally, discipline in the schools has focused on the use of punishment in an effort to control the misbehavior of specific students. Not only is punishment generally ineffective in achieving long-term suppression of problem behavior it does not help students learn desired, pro-social behaviors. Among the most important advances in student discipline procedures over the past decade is the development of school wide behavior support system that promotes and supports positive behaviors by all students. The goal of school wide systems are to define, teach and support appropriate behaviors in a way that enhances the academic and social behavior success of all students.

3.4 Autistic Spectrum Disorders

- **3.4.1 Definition of Autism Spectrum Disorder (ASD)**

- **a complex developmental condition that involves;**
- **persistent challenges in social interaction,**
- **speech and nonverbal communication, and**
- **restricted/repetitive behaviors.**
- **people with ASD may communicate, interact, behave, and learn in ways that are different from most other people.**
- **The learning, thinking, and problem-solving abilities of people with ASD can range from gifted to severely challenged.**
- **Some people with ASD need a lot of help in their daily lives; others need less.**
- **The effects of ASD and the severity of symptoms are different in each person.**

3.4.2 Signs and Symptoms Autism Spectrum Disorder (ASD):

- (1) Not point at objects to show interest (for example, not point at an airplane flying over).
- (2) Not look at objects when another person points at them
- (3) Have trouble relating to others or not have an interest in other people at all
- (4) Avoid eye contact and want to be alone
- (5) Have trouble understanding other people's feelings or talking about their own feelings
- (6) Prefer not to be held or cuddled, or might cuddle only when they want to
- (7) Appear to be unaware when people talk to them, but respond to other sounds
- (8) Be very interested in people, but not know how to talk, play, or relate to them
- (9) Repeat or echo words or phrases said to them, or repeat words or phrases in place of normal language
- (10) Have trouble expressing their needs using typical words or motions
- (11) Not play "pretend" games (for example, not pretend to "feed" a doll)
- (12) repeat actions over and over again
- (13) Have trouble adapting when a routine changes
- (14) Have unusual reactions to the way things smell, taste, look, feel, or sound
- (15) lose skills they once had (for example, stop saying words they were using)

3.4.3 Causes and Contributing Factors:

- **We do not know all of the causes of ASD. However, many causes for multiple types of ASD.**
- **It include;**
 - **environmental,**
 - **biologic and genetic factors. Individuals with certain genetic or**
 - **chromosomal conditions, such as fragile X syndrome or tuberous sclerosis,**
 - **the prescription drugs valproic acid and thalidomide have been linked with a higher risk of ASD.**
 - **critical period for developing ASD occurs before, during, and immediately after birth.**
 - **Children born to older parents are at greater risk for having ASD.**

3.4.4 Types of Autism

- There are five types of autism. These are:
- **(A) Level 1 Autism Spectrum Disorder.** Most people with this disorder are of normal or above-normal intelligence and possess strong verbal skills, but social communication is a challenge.
- **(B) Rett's Syndrome** affects girls and begins to become evident around 6 months old.
- Symptoms associated with Rett's Syndrome include ,
- social communication and
- an impaired ability to use one's hands (such as difficulty with gross and fine motor skills or repetitive hand and arm flapping),
- **(C) Childhood Disintegrative Disorder (CDD)** suddenly hit a wall and begin regressing around the age of two. This type of autism can seem especially crushing for parents as it often spawns confusion and fear.

- **(D) Kanner's Syndrome** This type of autism is also known as Classic Autistic Disorder, and its symptoms can include;
 - challenges communicating or understanding others,
 - engaging in virtually no eye contact, and
 - a hypersensitivity to stimuli (smell, light, noise, taste, or touch).
- Kanner's Syndrome display a profound need for routine and often display **no interest in the world around them**.
- These children turn their attention **inward and show little desire to interact with others**.
- **(E) Pervasive Developmental Disorder**
 - This form of autism is **generally less severe than others**.
 - have experienced delays in benchmarks, such **as speaking or walking, and**
 - **often lag behind other on pace to hit their developmental milestones**.
 - milder form more easily

3.4.5 Characteristics of Children with ASD

- **: Children with ASD often have/ difficulty with**
 - (1) social interaction**
 - (2) an unusual interest in objects**
 - (3) changes in routine**
 - (4) great ability in one area and great difficulty in another**
 - (5) may have unusually strong reactions to one or more of their five senses (6) may do the same thing over and over again, or talk constantly about specific things that interest them**
 - (7) unusually intense and prolonged emotional reactions and**
 - (8) while every child with ASD is unique, common characteristics include effects on the five senses, social interactions, and emotional expression**

Cont...

- These are also some of the characteristics of ASD:
- (1) problems with social interaction with others. This may include problems talking back and forth, working, or playing with others
- (2) need for sameness
- (3) great variation in abilities
- (4) under or over reaction to one or more of the five senses: sight, touch, taste, smell, or hearing
- (5) repeated actions or body movements and
- (6) unusual emotional reactions and expression. Children with ASD may have great ability in one area and great difficulty in another. But not all children with ASD have the same abilities.

3.4.6 Treatment and Intervention strategies for Children with Autism Spectrum Disorders:

- **A) Behavior and Communication Approaches help children with ASD are those that provide structure, direction, and organization for the child in addition to family participation.**
- **(B) Applied Behavior Analysis (ABA)**

Cont...

- There are different types of ABA. Here are some examples:
- **(1) Discrete Trial Training (DTT);**
- Lessons are broken down into their simplest parts, and **positive reinforcement is used to reward correct answers and behaviors. Incorrect answers are ignored;**
- **(2) Early Intensive Behavioral Intervention (EIBI);**
- EIBI uses a **highly structured teaching approach to build positive behaviors (such as social communication) and reduce unwanted behaviors (such as tantrums, aggression, and self-injury).**
- EIBI takes place in **a one-on-one adult-to-child environment** under the supervision of a trained professional.
- **(3) Early Start Denver Model (ESDM)** to help children advance their **social, language, and cognitive skills**. This can be done through **(1) Pivotal Response Training (PRT)** PRT aims to increase a child's motivation to learn, monitor their own behavior, and initiate communication with others. Positive changes in these behaviors are believed to have widespread effects on other behaviors. **(2) Verbal Behavior Intervention (VBI)** and **(3) VBI is a type of ABA that focuses on teaching verbal skills.**

Complete treatment program for a child with ASD:

- **Assistive Technology** such as **communication boards and electronic tablets**, can help people with ASD communicate and interact with others. For example, the **Picture Exchange Communication System (PECS)**
- **Developmental, Individual Differences, Relationship-Based Approach** (also called “**Floortime**”) focuses on emotional and relational development (feelings and relationships with caregivers). It also focuses on how the child deals with sights, sounds, and smells. TEACCH uses visual cues to teach skills.
- **Occupational Therapy** teaches skills that help the person live as independently as possible. Skills may include **dressing, eating, bathing, and relating to people**.
- **Social Skills Training** teaches children the skills they need to interact with others, including **conversation and problem-solving skills**.
- **Speech Therapy** helps to improve the person’s communication skills. Some people are able to learn verbal communication skills.
- **Dietary Approaches:** Some dietary treatments have been developed to address ASD symptoms.

The Treatment and Education of Autistic and Related Communication-Handicapped Children (TEACCH)

- The principles of TEACCH's structured teaching include:
 - (1) Understanding the culture of autism
 - (2) Developing an individual- and family-centered plan for each student, rather than using a standard curriculum
 - (3) Structuring the physical environment in a way that will assist students with autism to understand meaning
 - (4) Using visual supports to make the sequence of daily activities predictable and understandable and
 - (5) Using visual supports to make individual tasks understandable

UNIT FOUR: Gifted and Talented Children and Other Vulnerable Group

4.1 Gifted and Talented Children

- **4.1.1 Definition of Gifted and Talented Children**
- **persons whose abilities, talents, and potential for accomplishment are so exceptional or developmentally advanced that they require special provisions to meet their educational programming needs.**

4.1.2 Characteristics of Gifted and Talented Children

- **Cognitive**
- Keen power of abstraction
- Interest in problem-solving and applying concepts
- Voracious and early Reader
- Large vocabulary
- Intellectual curiosity
- Power of critical thinking, skepticism, self-criticism
- Persistent, goal-directed behavior
- Independence in work and study
- Diversity of interests and abilities

Creative

- ☐ Creativeness and inventiveness
- ☐ Keen sense of humor
- ☐ Ability for fantasy
- ☐ Openness to stimuli, wide interests
- ☐ Intuitiveness
- ☐ Flexibility
- ☐ Independence in attitude and social
- ☐ behavior
- ☐ Self-acceptance and unconcern for social norms
- ☐ Radicalism
- ☐ Aesthetic and moral commitment to self-selected work
- ☐

Affective

- Unusual emotional depth and intensity
- Sensitivity or empathy to the feelings of others
- High expectations of self and others, often
- leading to feelings of frustration
- Heightened self-awareness, accompanied by feelings of being different
- Easily wounded, need for emotional support
- Need for consistency between abstract
- values and personal actions
- Advanced levels of moral judgment
- Idealism and sense of justice

Behavioral

- Spontaneity
- Boundless enthusiasm
- Intensely focused on passions—resists changing activities when engrossed in own interests
- Highly energetic—needs little sleep or down time
- Constantly questions
- Insatiable curiosity
- Impulsive, eager and spirited
- Perseverance—strong determination in areas of importance
- High levels of frustration-particularly when
- having difficulty meeting standards of performance (either imposed by self or others)
- Volatile temper, especially related to perceptions of failure
- Non-stop talking/chattering

4.1.3 Supporting the gifted and talented

- **Acceleration** involves either a child's skipping a grade, or else the teacher's redesigning the curriculum within a particular grade or classroom so that more material is covered faster.
- **Enrichment** involves providing additional or different instruction added on to the usual curriculum goals and activities. Instead of books at more advanced reading levels,
- Obviously **acceleration and enrichment** can sometimes be combined. A student can skip a grade and also be introduced to interesting "extra" material at the new grade level. instruction.

4.2 Other Vulnerability Groups

- **4.2.1 Definition of Vulnerability** According to World Health Organization (WHO) and United Nations Children's Fund (UNICEF) the term “vulnerability” refers to **a condition of being weak or poorly defended or a condition in which an individual's basic needs of survival, safety, and development are jeopardized or threatened due to different factors.**
- Vulnerability can also be defined as someone, an adult or child, **is living in difficult circumstances, which may pose a threat to their physical, mental, emotional or social welfare.**

4.2.2 What makes children vulnerable?

- Social / Environmental related factors
- Poverty – Lack of food, Shelter,...
- Lack of psychosocial and emotional support
- Does not receive care, love, hugging (even for the rich)
- Experience abuse
- Live in a street
- Trafficked in to forced work
- Live in institutions
- Child labor
- Live in armed conflict area
- Refugee or displaced
- Exposure to crime, gangs, drugs use
- Early married, Child/unwanted pregnancy
- Social isolation (because of their gender, age, race....)
- Poor biological environment
- Poor access to school, health service, playground

Personal / Health related factors

- Chronic health problems
- Disability
- Emotional or psychological problems
- Sleeping disorder
- Drug abuse

Family related factors

- Death of parents/loved ones
- Caregivers who are alcoholic and unwilling to care
- Overcrowded house / large number of children
- Abusive/background of it, lack parenting knowledge
- Divorced / Single, parents in abusive relationship
- Chronically ill, disabled,
- Very old / young, non-biological adult at home
- Unemployed, Low income parents

4.2.3 Categories of Vulnerable Children

- **A) Street Children:** children with **no real home to go to, and "children on the street", who work or spend extensive time in public spaces, either alone or with relatives, guardians or other children, but have some sort of home.**
- **B) Orphan/s** is a child under the age of 18 who has either lost one or both parents” .
- **C) Children Live in Poverty:** Poverty can make children to be vulnerable to community shocks. **weaken the protection of children in every community.**

Cont...

- D) **Migrant Children:** due to **famine, persecution, religious intolerance, racial discrimination or any other reason**. Such people are **unable to exercise freely all the human rights**
- E) **Biologically at Risk Children** are at increased risk factors for developmental delay or later difficulties. include **prematurity and low birth weight**. People those who are of 60 years Very often, the rights of these people are **violated by family members, society and in other fronts. Being elders, their rights need to be protected.**
- F) **Children with Disabilities:** leads to the **loss or limitation of opportunities to take part in the life of the community on an equal level with others."** Sometimes a mild impairment can lead to more of a disability than a significant impairment, depending on the circumstances.

Cont...

- **G) Minority Groups:** Children from minority ethnic backgrounds are more likely to be unemployed as adults than the general population. However, children from minority ethnic backgrounds are also more likely to live in **persistent poverty, which in turn leads to worse outcomes.**
- **H) Children Affected by Armed Conflict:** Armed conflict can make children critically vulnerable in a multitude of ways. They may suffer indirectly as their parents' livelihood is affected by **market breakdowns caused by the conflict.** Vulnerability is particularly high who live in temporary camps that present a host of new risks, including **epidemics, insecurity, prostitution**, etc
- **I) Children *Exposed to Child Labor*:** the term child labor is used for employment of children below a certain age, which is considered illegal by law and custom. considered exploitative and inhuman by many international organizations.